Finite Observations, Infinite Behaviour: categorical semantics for stateful processes

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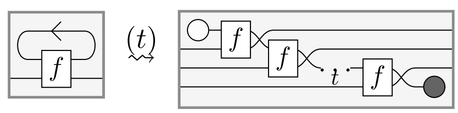
State is a fundamental concept in computer science which is used to model processes which evolve over space and time. Examples of stateful processes include automata, stream transducers, and imperative programs with mutable variables.

Stateful processes are well-understood for *classical deterministic total* processes. For example, *Mealy machines* are a simple class of stateful processes. A Mealy machine consists of a transition function $f : S \times X \longrightarrow S \times Y$ which at each time-step consumes an input in X, updates an internal state in S, and produces an output in Y [Mea55]. To run a Mealy machine for some finite amount of time, one must first initialise the state, repeatedly feed back the internal state to the transition function at the next time-step, and then discard the final state:



(1)

However, from this intensional description, it is not easy to tell when two Mealy machines will produce the same processes when they are run over time, that is to say, when they will have the same *behaviour*. One solution to this problem is to use coalgebraic techniques (see [Rut00] for reference). A Mealy machine $f : S \times X \to Y \times S$ and an initial state $s_0 \in S$ determine a stream transducer $\mathsf{run}(f, s_0) : X^{\mathbb{N}} \to Y^{\mathbb{N}}$ defined coinductively by the fixed point:

$$\mathsf{run}(f, s_0)(x_0, x_1, \dots) = \big(y_0, \mathsf{run}(f, s_1)(x_1, x_2, \dots)\big), \quad \text{where} \quad (y_0, s_1) = f(s_0, x_0).$$

Two Mealy machines f and g, with initial states s_0 and t_0 can be said to have the same *behaviour* over time when $\mathsf{run}(f, s_0) = \mathsf{run}(g, t_0)$.

Approximation. Despite its elegant formulation, the coalgebraic approach collapses in the presence of zero morphisms (Lemma 2.18) and compact closure (Lemma 2.17). These properties arise naturally across many settings. Quantum processes (completely positive maps, completely positive trace-nonincreasing maps), nondeterministic processes (relations), partial processes (partial functions), and partial probabilistic processes (subprobability kernels) all have zero morphisms. Moreover, completely positive maps and relations are also compact closed. To capture the behaviour of more general notions of stateful processes, we take inspiration from domain theory: we represent these processes not as fixed points in the above sense, but rather in terms of a series of *approximations* over time [Sco70].

In order to define a reasonable notion of approximation in a symmetric monoidal category, we weaken Carboni and Walter's notion of a bicategory of relations [CW]:

Definition (Definitions 2.20 and 4.4)**.** A **discard bicategory**[1] is a preorder-enriched symmetric monoidal category, such that for all objects X, there exists a **discard map** $\top_X : X \to I$, so that $\top_Y \circ f \subseteq \top_X$, $\top_I = 1_I$ and for all $f : X \to Y$ $\top_{X \otimes Y} = \top_X \otimes \top_Y$.

We interpret $f \subseteq g$ to denote that a morphism g approximates f. Furthermore, we interpret $\top_X$ as the process which forgets the information of the system on X.

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[1]We use the name "laxly natural discard category" in the draft.

Categorical semantics for stateful processes. Using this structure of a discard bicategory, we formalise what is meant by "approximations" in a symmetric monoidal category, following Carette et al. [CdVP21, Definition 18]:

Definition. A sequence of morphisms in a discard bicategory,

$$\{f_t : X_0 \otimes \cdots \otimes X_t \to Y_0 \otimes \cdots \otimes Y_t\}_{t\in\mathbb{N}},$$

is **monotone** in case for all $t \in \mathbb{N}$, $(1_{Y_0\otimes\cdots\otimes Y_t} \otimes \top_{Y_{t+1}}) \circ f_{t+1} \subseteq f_t \otimes \top_{X_{t+1}}$.

We interpret the element f_t of the sequence $\mathbf{f}$ to denote the knowledge or constraints about the stateful system which can be observed at time t. Under this view, monotonicity imposes that progressing to the next time step $t + 1$ and forgetting the new output can only reveal additional constraints about the system at time t. This cannot reveal contradictory information, nor can it forget information from the previous time-step.

Carette et al. previously used monotone sequence to model unnormalised stateful quantum processes [CdVP21]. If we interpret each approximation as incomplete information about the underlying stateful process, we can view the behaviour of a process as the collection of constraints which will be revealed at any point in time. Under this interpretation, a zero morphism, represents inconsistent constraints. Indeed, unlike the coalgebraic approach, monotone sequences do not collapse when the base category has zero morphisms. However, monotone sequences fails to equate proceses which ought to have the same behaviour in this sense. For example, a monotone sequence which imposes inconsistent constraints at time t will not in general be equivalent to one which is known to be inconsistent only at time $t + 1$.

We provide a mathematically elegant and simple solution to this problem: we assert that two sequences have the same behaviour in case they can be seen to approximate each other:

Definition (Definition 3.3)**.** Given two monotone sequences $\mathbf{f}, \mathbf{g} : \mathbf{X} \to \mathbf{Y}$, say that $\mathbf{g}$ ***approximates*** $\mathbf{f}$, written $\mathbf{f} \subseteq \mathbf{g}$, if for all $n \in \mathbb{N}$ there exists some **lookahead** $m \in \mathbb{N}$ such that:

$$(1_{Y_0\otimes\cdots\otimes Y_n} \otimes \top_{Y_{n+1}\otimes\cdots\otimes Y_{n+m}}) \circ f_{n+m} \;\subseteq\; g_n \otimes \top_{X_{n+1}\otimes\cdots\otimes X_{n+m}}.$$

Two monotone sequences are **observationally equivalent** in case they approximate each other.

In other words, two sequences are observationally equivalent if at some point in time, the information which is encapsulated by the one process will later be revealed by the other, and vice versa. This leads to category whose morphisms are the behaviours of stateful processes:

Definition (Definition 4.1)**.** Given a discard bicategory $\mathcal{C}$, the category $\mathrm{Obs}_{\mathbb{N}}(\mathcal{C})$ of initialised **observational behaviours** has objects given by infinite sequences of objects in $\mathcal{C}$ and morphisms given by observational equivalence classes of monotone sequences.

This construction preserves the structure which we used to define approximation:

Theorem (Theorem 3.7)**.** *$\mathrm{Obs}_{\mathbb{N}}(\mathcal{C})$ is a discard bicategory.*

Unlike monotone sequences, we show that all inconsistent behaviours are equivalent. Moreover, by contrast to the coalgebraic approach, just as for monotone sequences, our construction does not collapse when applied to arbitrary compact closed categories.

We use this machinery to prove various results. For example, we prove a categorified compactness theorem for nondeterminstic processes. If we restrict ourself to the category $\mathsf{Rel}(\mathsf{KHaus})$ of closed relations between compact Hausdorff spaces, behaviours extend to a unique infinite relation:

Theorem (Theorem 5.5)**.** *There is a discard bicategory functor* $\mathrm{Obs}_{\mathbb{N}}(\mathsf{Rel}(\mathsf{KHaus})) \to \mathsf{Rel}(\mathsf{KHaus})$.

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Abstract

Time-dependent processes are often described in terms of machines with an internal state which is updated as time evolves. However, from the perspective of an external observer, two such processes can produce the same behaviour, even when their internal states differ. We introduce a semantic construction which equates processes when they exhibit the same infinite external behaviour, as determined by all finite observations. Our construction is defined over any preorder-enriched monoidal category with discarding, providing a unified framework for partial, non-deterministic, probabilistic, and quantum processes evolving in space and time. In all these cases, our construction detects failures and supports a delayed trace interpretation of stateful machines. For non-deterministic systems, we establish a compactness theorem: under suitable topological assumptions, every family of compatible finite observations extends uniquely to a coherent infinite behaviour. Finally, we investigate linear time-invariant systems, relating our semantics to the signal flow graphs of Bonchi et al. via an oplax functor that captures the inverse z-transform, and showing that our construction applied to linear relations recovers Willems' notion of behaviour.

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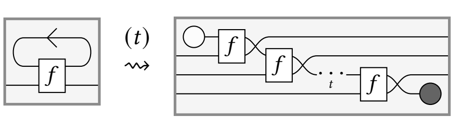
1 Introduction

Many computational systems of interest in science and engineering evolve by processing an infinite stream of information: automata [41, 43], petri nets [49], signal flow graphs [55], stochastic processes [22], and quantum channels with memory [35].

A common specification of such a process is a *Mealy machine*: a transition function $f : S \otimes X \longrightarrow S \otimes Y$ which at each time-step consumes an input in X, updates an internal state in S, and produces an output in Y [41]. Its behaviour over time is obtained by initialising the state, repeatedly feeding back the internal state to the transition function and then discarding the final state:



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However, two Mealy machines may have different internal state spaces and transition structures, yet induce the same input–output behaviour. This raises the fundamental question:

When do two stateful machines produce the same infinite behaviour?

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Following the tradition initiated by Scott [53] and later developed by Plotkin [51] as well as Abramsky and Jung [2], we seek a mathematical answer to this question by means of denotational semantics: rather than describing systems operationally, we aim to interpret them as mathematical objects whose equality precisely captures behavioural equivalence.

Our approach is process theoretic: we interpret machines as morphisms in symmetric monoidal categories [18]. Different choices of monoidal categories allow us to model a wide range of computational phenomena, including partiality [17], nondeterminism [13], probabilistic [23], and quantum computation [1]. From this perspective, we aim to answer the question above systematically, in a way that applies uniformly to all of these theories of processes.

Building on Sabadini and Walter's notion of a feedback category [33], recent work has made substantial progress in developing semantic frameworks for stateful systems [6, 14, 19, 56]. Particularly in *total* deterministic and stochastic settings, coalgebraic methods provide elegant characterisations of infinite behaviour [20]. However, when one moves beyond totality to *partial*, *nondeterministic*, or *quantum* processes, these approaches encounter serious difficulties: failures may not propagate correctly, compact structure can trivialise infinite behaviour, the time-delay may not be natural, and different semantic constructions can either identify too much or too little.

In this paper, we return to the insights of Scott by reframing the problem in terms of *finite approximations*. Our key observation is that many of the above difficulties disappear once we work not merely in monoidal categories, but in *preorder-enriched* monoidal categories, where morphisms carry an intrinsic notion of approximation or information content. This enrichment allows us to interpret finite observations as partial information about an infinite behaviour, ordered by refinement.

> "Suppose or $x, y \in D$ are two elements of the data type, then the idea is not immediately to think of them as being completely separate entities just because they may be *different*. Instead y, say, may be a better version of what x is trying to approximate. In fact, let us write the relationship $x \sqsubseteq y$ to mean intuitively that y is consistent with x and is (possibly) *more accurate* than x. For short we can say that x *approximates* y."
>
> (Scott, 1970) [53]

Building on this idea, we introduce a semantic construction in which the infinite behaviour of a process is determined by the totality of its finite observations, subject to the condition that larger observations can only refine smaller ones. Two stateful processes are identified when they can be seen to approximate each other. The resulting framework applies uniformly to partial, nondeterministic, probabilistic, and quantum processes: providing a denotational semantics that detects failures, admits a natural delayed trace, and captures infinite behaviour via finite approximation.

Contributions:

- We construct a poset-enriched monoidal category of observational behaviours $\mathrm{Obs}_{I,\mathcal{J}}(C)$ (Definition 3.6 and Theorem 3.7) over any preorder-enriched discard category C, parameterised by a space-time index set I and a directed family of observable finite contexts $\mathcal{J}$. Unlike previous approaches, our construction detects failures (Proposition 3.9) and applies uniformly across partial, non-deterministic, stochastic and quantum processes.
- We show that, under a mild assumption that holds across our examples, *initialised* stateful machines admit a functorial interpretation as behaviours compatible with feedback structure (Theorem 4.4), which is moreover universal for both classical (Proposition 4.5) and quantum (Proposition 4.7) processes. In the compact-closed setting, we introduce a feedback compact-closed category of *uninitialised* behaviours (Theorem 4.10), which detects failures and admits a canonical feedback-preserving interpretation of the State construction (Corollary 4.11).
- We prove a nondeterministic, functorial compactness theorem for closed relations between compact Hausdorff spaces, showing in this setting that local finite approximations uniquely determine a global process (Theorem 5.5). Moreover, we show that this functor is faithful for relations between finite sets (Corollary 5.6).
- We exhibit an oplax feedback functor from Bonchi et al.'s frequency domain semantics for linear time-invariant systems to observational behaviours (Theorem 6.2). Our construction provides the first categorical semantics for linear time-invariant systems in the time-domain. We use the compactness theorem to show that our notion of behaviours coincides with that of Willems for finite fields (Theorem 6.5). We end by discussing the syntactic implications for signal flow graphs (Proposition 6.9).

2 Preliminaries

We review the instantiation of Mealy machines in monoidal categories and their semantics in terms of stateful morphisms, coalgebraic streams and causal processes. We highlight the limits of these approaches in examples of interest: partial, nondeterministic, probabilistic and quantum processes.

2.1 Mealy machines in monoidal categories

Mealy machines [41] are a common description of stateful processes consisting in a transition function $S \times X \to S \times Y$ taking inputs in X and giving outputs in Y while updating an internal state in S. In Cartesian closed categories, stateful processes are often packaged using the *state monad* $T_S(X) \coloneqq (X \times S)^S$ [42]. A Kleisli arrow for the state monad is precisely a Mealy machine and iterating the Kleisli composition gives the usual n-step time evolution of the machine (see Equation (1)).

In this paper, we are interested in a generalised notion of Mealy machine where the underlying transition $S \times X \to S \times Y$ is not necessarily a function, or even a total deterministic process. For example, we are interested in machines built out of partial, non-deterministic, probabilistic or quantum processes. These live in symmetric monoidal categories, which formalise process theories with a reasonable notion of parallel and sequential composition.

Definition 2.1. A **strict symmetric monoidal category** (SMC) is a category C equipped with a functor $\otimes : C \times C \to C$, called the **monoidal product**, and an object $I \in C$, called the **monoidal unit**. For all objects X, Y, Z, we impose that:

$$(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z), \quad \text{and} \quad I \otimes X = X = X \otimes I,$$

in addition to a natural isomorphism, the **symmetry**:

$$\mathsf{swap}_{X,Y} : X \otimes Y \to Y \otimes X \quad \text{such that} \quad \mathsf{swap}_{X,Y} = \mathsf{swap}^{-1}_{Y,X}.$$

A **compact-closed category** (CCC) is a SMC where every object X has a *dual* X^*, such that for all objects X and Y, $(X \otimes Y)^* = Y^* \otimes X^*$; for all objects X, there are morphisms

$$\cup_X : I \to X \otimes X^*, \quad \text{and} \quad \cap_X : X^* \otimes X \to I,$$

such that the following equations hold

$$(1_X \otimes \cap_X) \circ (\cup_X \otimes 1_X) = 1_X, \quad (\cap_X \otimes 1_{X^*}) \circ (1_{X^*} \otimes \cup_X) = 1_{X^*}.$$

Non-strict SMCs are defined by replacing the equalities above with coherent natural isomorphisms. Since every SMC is symmetric monoidally equivalent to a strict SMC [28], we can work in the strict setting without loss of generality.

The concept of Mealy machine can be generalised to any SMC:

Definition 2.2. Let C be a symmetric monoidal category and let S be an object of C. A **symmetric monoidal Mealy machine** is a morphism $f : S \otimes X \to S \otimes Y$, where X is interpreted as the input, Y as the output and S as the internal state. For $n \geq 1$, its n-step evolution is the canonical composite $f^{(n)} : S \otimes X^{\otimes n} \longrightarrow Y^{\otimes n} \otimes S$ obtained by wiring together n copies of f so that the output state of the k-th copy is fed back to the input state of the $(k+1)$-st copy, repeatedly using swap, as in Equation (1).

Given an **initial state** $s_0 : I \to S$, the **evaluation** at time-step n is given by $f^{(n)} \circ (s_0 \otimes 1_{X^{\otimes n}}) : X^{\otimes n} \to Y^{\otimes n} \otimes S$.

To illustrate the above definitions, we now give the main examples which will be used throughout the paper.

2.1.1 Classical processes. First, we shall review several different notions of classical processes which exist in the literature. We regard all the classical notions of processes theories as SMCs with respect to the Cartesian product (or the obvious analogues of the Cartesian product).

Example 2.3 (Partial processes). The category Par of sets and partial functions has sets as objects. Morphisms $f : X \to Y$ consist of a subset $X_f \subseteq X$, the *domain of definition*, as well as a function $f : X_f \to Y$. Given inputs $x \in X \backslash X_f$ which are not in the domain of definition, we say that the function f *fails* at x, i.e. it does not return any output.

Par is a canonical example of Robinson and Rossellini's notion of a *p-category*, which we shall refer to as a **Cartesian restriction category** following Cockett and Lack [17, P. 21]. We interpret Mealy machines over Cartesian restriction categories as automata which can possibly fail on certain inputs.

Example 2.4 (Nondeterministic processes). Nondeterministic processes are commonly modelled in the category Rel of sets and relations. Here, the objects are sets, morphisms $R : X \to Y$ are subsets $R \subseteq X \times Y$. Composition and identies are given by

$$R \circ S \coloneqq \{(x, z) \mid \exists y : (x, y) \in S, (y, z) \in R\}, \text{ and } 1_X \coloneqq \{(x, x) \in X^2\}.$$

Rel is the canonical concrete setting in which to model nondeterministic processes, because unlike Set or Par, inputs can be mapped to *multiple* number of outputs and vice-versa. More generally, given a regular category C we can form the category of *internal relations*, denoted $\mathsf{Rel}(C)$; so that in particular, $\mathsf{Rel}(\mathsf{Set}) \cong \mathsf{Rel}$.

Relations internal finite dimensional vector spaces are a categorical semantics for linear control theory [4, 7–11], as we will investigate further in Section 6. Linear relations also give categorical semantics for mechanical networks such as electrical circuits [3, 5].

Categories of internal relations are canonical examples of Carboni and Walters' notion of a **Cartesian bicategory** [13, § 1.], which are necessarily compact-closed. We interpret Mealy machines over Cartesian bicategories as nondeterministic automata.

Example 2.5 (Probabilistic processes). Functions with random outcomes are commonly modelled in the category of measurable spaces and probability kernels [24]. We will mostly consider the subcategory BorelStoch where the objects are standard Borel spaces (X, Σ_X) where Σ_X is the Borel σ-algebra on a Polish space X. The morphisms $f : (X, \Sigma_X) \to (Y, \Sigma_Y)$ are Markov kernels; i.e. a function $f : X \times \Sigma_Y \to [0, 1]$ which is both measurable in X and a probability measure over Σ_Y. For discrete probability theory, one may also consider the full subcategory FinStoch $\rightarrowtail$ BorelStoch where objects are finite sets and morphisms are stochastic matrices.

The category BorelStoch is the canonical example of a **Markov category with conditionals** [23, Ex. 11.3.]. Markov categories with conditionals provide a rich categorical framework for probability theory. We interpret Mealy machines over Markov categories with conditionals as probabilistic automata with a random transition functions, such that their time evolution generates controlled stochastic processes evolving over time.

Example 2.6 (Partial probabilistic processes). We will also consider categories of partial probabilistic processes, such as Panangaden's category of probabilistic relations or subprobability kernels [48, Def. 3.1], which corresponds to the Kleisli category of the composition of the Maybe monad and the Giry monad. The well-behaved subcategory $\mathsf{BorelStoch}_{\leq 1}$ is defined in the same way as BorelStoch except that $f : X \times \Sigma_Y \to [0, 1]$ is allowed to be a subprobability measure over Σ_Y, i.e. for all $x \in X$, $f(x, Y) \leq 1$.

$\mathsf{BorelStoch}_{\leq 1}$ is the canonical example of Di Lavore et al.'s notion of a *partial Markov category* (with conditionals) [37, Cor. 3.13.] which allow modelling updating and conditioning in substochastic processes.

All of the examples reviewed so far: Cartesian restriction categories, Cartesian bicategories, and Markov categories with conditionals, are indeed partial Markov categories [21, Props. 2.13, 2.16]. Therefore, we recall the explicit definition:

Definition 2.7. A **partial Markov category** is a symmetric monoidal category where each object X is equipped with a *copy map* $\delta_X : X \to X \otimes X$ and a *discarding map* $\top_X : X \to I$ such that:

- $(\delta_X, \top_X)$ is a cocommutative comonoid;
- comultiplication is uniform, so that for all objects X and Y

$$\delta_{X \otimes Y} = (1_X \otimes \mathsf{swap}_{X,Y} \otimes 1_Y) \circ (\delta_X \otimes \delta_Y), \quad \delta_I = 1_I;$$

- discarding is uniform, i.e. $\top_{X \otimes Y} = \top_X \otimes \top_Y$ and $\top_I = 1_I$;
- every morphism $f : X \to Y \otimes Z$ has a *conditional*, i.e. there exists some (possibly non-unique) $c : Y \otimes X \to Z$ such that

$$f = (1_Y \otimes c) \circ (\delta_Y \otimes 1_Z) \circ ((1_Y \otimes \top_Z) \circ f) \otimes 1_X) \circ \delta_X.$$

2.1.2 Quantum processes. In this paper we are also interested in stateful quantum processes. In particular, we will focus on the following classes of processes used in quantum information theory (see Nielsen and Chuang [44] for reference). Given a finite-dimensional Hilbert space $\mathcal{H}$, let $\mathcal{B}(\mathcal{H})$ be the space of linear operators on $\mathcal{H}$. Write $0 \preceq a$ to mean that a is *positive semidefinite*, i.e. $a = b^\dagger b$ for some $b \in \mathcal{B}(\mathcal{H})$. A linear map $\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{K})$ is:

(1) *positive* if for every $a \in \mathcal{B}(\mathcal{H})$, $0 \preceq a$ implies $0 \preceq \Phi(a)$;
(2) *completely positive* if for all finite-dimensional Hilbert spaces $\mathcal{E}$, the map $\Phi \otimes \mathrm{id}_{\mathcal{B}(\mathcal{E})}$ is positive;
(3) *trace-preserving* if for all $a \in \mathcal{B}(\mathcal{H})$, $\mathrm{Tr}(\Phi(a)) = \mathrm{Tr}(a)$;
(4) *trace-nonincreasing* if for all $a \in \mathcal{B}(\mathcal{H})$ with $0 \preceq a$,

$$\mathrm{Tr}(\Phi(a)) \leq \mathrm{Tr}(a).$$

Here Tr denotes the usual linear algebraic trace.

Define the categories CPM, CPTP and CPTNI whose objects are of the form $\mathcal{B}(\mathcal{H})$, for finite-dimensional Hilbert spaces $\mathcal{H}$; and whose morphisms are respectively completely positive; both completely positive and trace-preserving; and both completely positive and trace-nonincreasing maps. In all cases, the symmetric monoidal structure is given by the bilinear tensor product and $\mathbb{C}$. Note that CPM is compact-closed; whereas CPTP and CPTNI are not.

Interpreting $\mathcal{B}(\mathcal{H})$ as the space of unnormalised quantum states on $\mathcal{H}$, CPM captures the unnormalised quantum processes; CPTP captures the physically implementable processes, also known as *quantum channels*; and CPTNI captures the post-selected processes, also known as *quantum operations*.

We interpret Mealy machines in CPTP as *quantum channels with finite memory*, following [35]. Quantum channels with memory are widely studied in the contexts of spin chains, quantum coding, and quantum communication [15, 46, 50]. Similarly to the classical, probabilistic setting, working over CPTNI or CPM allows to further capture conditioning and post-selection in these processes.

2.2 Stateful morphisms

It is natural to ask: *what is the categorical semantics of Mealy machines over an arbitrary symmetric monoidal category?* At a minimum, such a semantics should provide a principled way to *hide* the internal state S, turning a stateful description into an external process from X to Y as in Equation (1). Following the approach of Katis, Sabadini, and Walters [30, 31] (and later by various others such as [26, 56]), we axiomatise this state-hiding operation with the notion of a *feedback category*. Our formulation is a mild variant of that of Katis, Sabadini, and Walters [30], adapted to the guarded/delayed setting needed later:[1]

[1] More generally one could replace Γ, Δ and $\mathcal{D}$ with a monoidal endoprofunctor on C, modifying the sliding equation appropriately.

Definition 2.8. A **feedback category** is a tuple $(\mathcal{C}, \mathcal{D}, \Gamma, \Delta, \mathsf{fbk})$ where:

- $\Gamma : \mathcal{D} \to \mathcal{C}$ is a symmetric monoidal functor, called the **guard functor**;
- $\Delta : \mathcal{D} \to \mathcal{D}$ is a symmetric monoidal endofunctor, called the **delay endofunctor**;
- $\mathsf{fbk}_S : \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y) \to \mathcal{C}(X, Y)$ is a family of functions, called the **feedback operator**,

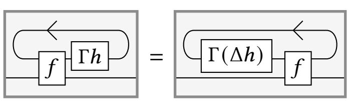
such that for all $f \in \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y)$, $u \in \mathcal{C}(X', X)$, $v \in \mathcal{C}(Y, Y')$, and $h \in \mathcal{D}(S, T)$ the following equations hold:

(1) **Tightening**
$$v \circ \mathsf{fbk}_S(f) \circ u = \mathsf{fbk}_S((1_{\Gamma S} \otimes v) \circ f \circ (1_{\Gamma(\Delta S)} \otimes u));$$
(2) **Vanishing** $\mathsf{fbk}_I(f) = f$;
(3) **Joining** $\mathsf{fbk}_T(\mathsf{fbk}_S(f)) = \mathsf{fbk}_{S \otimes T}(f)$;
(4) **Strength** $\mathsf{fbk}_S(f) \otimes g = \mathsf{fbk}_S(f \otimes g)$;
(5) **Sliding**
$$\mathsf{fbk}_T\big((\Gamma(h) \otimes 1_Y) \circ f\big) = \mathsf{fbk}_S\big(f \circ (\Gamma(\Delta h) \otimes 1_X)\big).$$

In case Γ is faithful call $\mathcal{D}$ the **guarded subcategory**.

A **feedback functor** between feedback categories
$$(F, F|_{\mathcal{D}}) : (\mathcal{C}, \mathcal{D}, \Gamma, \Delta, \mathsf{fbk}) \to (\mathcal{C}', \mathcal{D}', \Gamma', \Delta', \mathsf{fbk}')$$
consists of symmetric monoidal functors $F : \mathcal{C} \to \mathcal{C}'$ and $F|_{\mathcal{D}} : \mathcal{D} \to \mathcal{D}'$, satisfying $\Gamma' \circ F|_{\mathcal{D}} = F \circ \Gamma$, $\Delta' \circ F|_{\mathcal{D}} = F|_{\mathcal{D}} \circ \Delta$, and $F(\mathsf{fbk}_S(f)) = \mathsf{fbk}'_{F(S)}(F(f))$.

We interpret $\Gamma : \mathcal{D} \to \mathcal{C}$ as parametrising what is allowed to be passed through the state at successive timesteps, and $\Delta : \mathcal{D} \to \mathcal{D}$ as an effect which is applied when something is passed through the state; graphically:



(2)

In particular, when $\Gamma = \Delta = 1_{\mathcal{C}}$, a morphism $f : S \otimes X \to S \otimes Y$ is exactly a symmetric monoidal Mealy machine, where $\mathsf{fbk}_S(f) : X \to Y$ is interpreted as the process given by hiding the memory of f as an internal state.

Note that a *traced symmetric monoidal structure* [29] on a SMC $\mathcal{C}$ is precisely a feedback structure with $\Delta = \Gamma = 1_{\mathcal{C}}$ satisfying the additional *yanking identity*: $\mathrm{Tr}_X(\mathsf{swap}_{X,X}) = 1_X$, which trivialises the time dependency.

Generalising the "Circ" construction of Katis et al. [30], now more commonly referred to in the theoretical computer science literature as the *state construction* (see for example [19]), we can *freely* construct a feedback category to give a categorical semantics for stateful processes over any base symmetric monoidal category:

Definition 2.9 (State construction)**.** Given SMCs $\mathcal{C}$ and $\mathcal{D}$ with symmetric monoidal functors $\Gamma : \mathcal{D} \to \mathcal{C}$ and $\Delta : \mathcal{D} \to \mathcal{D}$, the SMC $\mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ has the same objects as $\mathcal{C}$. The morphisms given by elements of the set:
$$\mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(X, Y) \coloneqq \int^{S \in \mathcal{D}} \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y).$$
The rest of the structure defining a symmetric monoidal category is given by the evident natural transformations.

This integral symbol denotes a coend [38], so that elements of $\mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(X, Y)$ are given by pairs
$$(S \in \mathrm{ob}(\mathcal{D}), f \in \mathcal{C}(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y)),$$
modulo the largest equivalence relation such that for any $r : S \to S'$ and any $h : S' \otimes X \to S \otimes Y$:
$$[S', (\Gamma(s) \otimes 1_Y) \circ h] = [S, h \circ (\Gamma(\Delta(s)) \otimes 1_X)].$$

A morphism in $\mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ is called **stateful morphism**. In case, $\Gamma = 1_{\mathcal{C}}$ or $\Delta = 1_{\mathcal{D}}$, we omit the labels $\mathcal{D}$ or Δ.

The state construction is the *free* feedback category built from $\mathcal{C}$, $\mathcal{D}$, Γ, and Δ, see Proposition C.2 for a detailed proof. Given $f \in \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(\Gamma(\Delta S) \otimes X, \Gamma(S) \otimes Y)$, then $\mathsf{fbk}(f) : X \to Y$ is constructed by adding the object S to the memory. The definition of the coend means precisely that two stateful machines are equivalent when they are related by a finite number of applications of the sliding rule Equation (2).

We call the map $\delta_X \coloneqq \mathsf{fbk}_X(\mathsf{swap}_{\Gamma(\Delta X), \Gamma X})$, the **delay morphism** at X. Generalising the work of Katis et al. [31, P. 5], we find that given some weak assumptions on the guard functor, feedback structures in compact-closed categories are determined by the delay morphisms $\{\delta_X\}_{X \in \mathrm{ob}(\mathcal{D})}$:

Proposition 2.10. *Take a compact-closed category $\mathcal{C}$, a faithful symmetric monoidal functor $\Gamma : \mathcal{D} \rightarrowtail \mathcal{C}$ such that for all $X \in \mathrm{ob}(\mathcal{D})$, $\Gamma^{-1}(\delta_X) \neq \varnothing$, and a symmetric monoidal endofunctor $\Delta : \mathcal{D} \to \mathcal{D}$. The existence of a feedback operator is equivalent to a monoidal natural transformation $\delta : 1_{\mathcal{D}} \Rightarrow \Delta$, called the **delay natural transformation**. Moreover, such a natural transformation is necessarily a natural isomorphism.*

See proof on page 16. Applying the state construction to $\mathcal{C}$ identifies stateful processes build from $\mathcal{C}$ that differ only by a reparametrisation of the internal state. However, this semantics is purely formal: it does not specify how to iterate a process through time. To model iteration explicitly, we pass from $\mathcal{C}$ to categories of time-indexed families in $\mathcal{C}$. The key extra ingredient is a canonical endofunctor expressing a one timestep delay:

Definition 2.11. Let $\mathbb{Z}$ and $\mathbb{N}$ denote the discrete symmetric monoidal categories whose objects are respectively the integers and the natural numbers, where the monoidal product and unit are given by addition and 0.

Given a symmetric monoidal category $\mathcal{C}$, the functor categories $[\mathbb{Z}, \mathcal{C}]$ and $[\mathbb{N}, \mathcal{C}]$ have objects and morphisms given by (bi)infinite sequences of objects and morphisms in $\mathcal{C}$. Equipping these functor categories with the *pointwise* symmetric monoidal structure, we find that there are symmetric monoidal functors:
$$\begin{aligned} \mathtt{Shft} : [\mathbb{Z}, \mathcal{C}] \to [\mathbb{Z}, \mathcal{C}]; &\quad (\mathtt{Shft}\, X)_{j+1} \coloneqq X_j, \quad (\mathtt{Shft}\, f)_{j+1} \coloneqq f_j; \\ \Delta : [\mathbb{N}, \mathcal{C}] \to [\mathbb{N}, \mathcal{C}]; &\quad (\Delta X)_0 \coloneqq I, \quad (\Delta X)_{j+1} \coloneqq X_j, \\ &\quad (\Delta f)_0 \coloneqq 1_I, \quad (\Delta f)_{j+1} \coloneqq f_j. \end{aligned}$$
With this data, we can build feedback categories of **extensional monoidal streams** [19, § 3.2]. Call a morphism in $\mathrm{St}([\mathbb{N}, \mathcal{C}], \Delta)$ an **initialised stateful morphism sequence** in $\mathcal{C}$; and a morphism in $\mathrm{St}([\mathbb{Z}, \mathcal{C}], \mathtt{Shft})$ an **uninitialised stateful morphism sequence**.

A guard functor $\Gamma : \mathcal{D} \to \mathcal{C}$ induces functors $\Gamma : [\mathbb{N}, \mathcal{D}] \to [\mathbb{N}, \mathcal{C}]$ and $\Gamma : [\mathbb{Z}, \mathcal{D}] \to [\mathbb{Z}, \mathcal{C}]$ which can be used to restrict the sliding rule in stateful morphism sequences. For instance, initialised stateful morphism sequences with sliding guarded by Γ live in $\mathrm{St}_{[\mathbb{N},\mathcal{D}]}([\mathbb{N},C],\Delta)$. We however do not require this restriction in our results.

In the uninitialised setting, we can unroll stateful morphisms from the free feedback category into stateful morphism sequences analagously to Equation (1):

Lemma 2.12. *For every SMC C, there is a unique feedback functor $U : \mathrm{St}(C) \to \mathrm{St}([\mathbb{Z},C],\mathtt{Shft})$ sending:*

$$\Big([S,f] : X \to Y\Big) \;\mapsto\; \Big(\big[(f : S\otimes X \to S\otimes Y)_{j\in\mathbb{Z}}\big] : X^{\mathbb{Z}} \to Y^{\mathbb{Z}}\Big).$$

See proof on page 17.

Remark 2.13. On the other hand, unless C is a *monoid*, there is no feedback functor $F : \mathrm{St}(C) \to \mathrm{St}([\mathbb{N},C],\Delta)$ acting on objects by $X \mapsto (X,X,\dots)$, since we would then have that $(X,X,\dots) = F(1_C(X)) = \Delta(F(X)) = (I,X,X,\dots)$, and thus $I = X$ for all objects X. Nevertheless, one may still obtain an $\mathbb{N}$-indexed interpretation of stateful morphisms by equipping morphisms in St with an initial state as in the work of Bonchi et al. [6, § III.E.], but this requires changing the definition of the feedback operation [6, Def. III.15]. Here, we will use $\mathrm{St}([\mathbb{N},C],\Delta)$ directly as a syntax for initialised automata for convenience.

Extensional streams are a categorical semantics for Mealy machines, albeit in a different sense than for the state construction over the base category. Extensional streams identify processes where the state at any time can be propagate finitely far into the future or the past. Therefore the functor from Lemma 2.12 can be seen as adding a notion of discrete time dynamics to the more primitive categorical semantics described in the previous subsection.

There are some problems with extensional streams; for example, consider the following structure which we will use to identify that a failure has occurred during computation:

Definition 2.14. A SMC C has **monoidal zero morphisms** if for all objects X, Y there exists a morphism $0_{X,Y} : X \to Y$ such that for all morphisms $f : X' \to X$ and $g : Y \to Y'$,

$$0_{X,Y} \circ f = 0_{X',Y},\; g \circ 0_{X,Y} = 0_{X,Y'},\; \text{and } 1_Z \otimes 0_{X,Y} = 0_{Z\otimes X, Z\otimes Y}.$$

Example 2.15. Several of our examples have this property:

- Par has monoidal zero morphisms given by the functions with empty domains of definition.
- Rel has monoidal morphisms given by empty subsets.
- $\mathsf{BorelStoch}_{\leq 1}$ has monoidal zero morphisms given by the everywhere-zero kernels.
- CPM, and CPTNI have monoidal zero morphisms given by the zero linear maps.

If at some point in time we find locally that the process has failed, then intuitively we want to interpret the global process itself as having failed. However, due to the fact that extensional streams only allow for morphisms to slide *finitely* far into the past or future, it follows immediately that:

Lemma 2.16. *Take a symmetric monoidal category C with monoidal zero morphisms. Consider a representative dinatural stream $[\mathbf{f}]$ in $\mathrm{St}([\mathbb{Z},C],\mathtt{Shft})$ or $\mathrm{St}([\mathbb{N},C],\Delta)$ with some $f_j = 0$. Then in general $[\mathbf{f}]$ will not be a monoidal zero morphism.*

This is only one example of the many shortcoming of extensional streams. For example, by applying the feedback operator to an entire endomorphism, processes become hidden in the state; where extensionally indistinguishable processes may fail to be in the same equivalence class. However many of these problems are resolved via taking a more sophisticated coalgebraic approach.

2.3 Coalgebraic streams

Extensional streams only allow for the internal state at one time to be reparametrised finitely far into the past or the future. Therefore, in a certain sense, the induced equivalence relation is very coarse as it fails to account for truly infinite behaviour. In order to give a properly infinite stream semantics, in this subsection, we recall the coalgebraic notion of monoidal streams first developed by Di Lavore et al. [19, Def. 4.1] and later generalised by Bonchi et al. [6, Def. IV.4.].

Given a symmetric monoidal category C and a symmetric monoidal subcategory $\Gamma : \mathcal{D} \rightarrowtail C$, Bonchi et al. [6] define a feedback category, which we denote by $\mathrm{Stream}_{\mathcal{D}}(C)$, whose objects are infinite lists of objects in C, where the hom-sets are given by the final fixpoint of profunctors:[2]

$$\mathrm{Stream}_{\mathcal{D}}(C)(\mathbf{X},\mathbf{Y}) \cong \int^{M\in\mathcal{D}} C\big(X_0, (\Gamma M)\otimes Y_0\big) \times \mathrm{Stream}_{\mathcal{D}}(C)\big((\Gamma M)\cdot \mathbf{X}^+, \mathbf{Y}^+\big), \tag{3}$$

where given $\mathbf{X},\mathbf{Y} \in \mathrm{ob}(C)^{\mathbb{N}}$ and $N \in \mathrm{ob}(C)$, we denote

$$N\cdot\mathbf{X} \coloneqq (N\otimes X_0, X_1, X_2,\dots), \quad \text{and} \quad \mathbf{X}^+ \coloneqq (X_1, X_2, \dots).$$

In case $\Gamma = 1_C$ denote this category by $\mathrm{Stream}(C)$ as in [19]. The category $\mathrm{Stream}(C)$ naturally captures total stateful processes; for example, stateful processes built from Cartesian monoidal categories or Markov categories with ranges and conditionals [19, Rem. 4.10]. However, we argue in the presence of compact closure or monoidal zero morphisms, which are often present in nondeterministic partial and quantum settings, coalgebraic monoidal streams are poorly behaved.

In the presence of compact closure, coalgebraic monoidal streams without a guarded subcategory are degenerate:

Lemma 2.17. *$\mathrm{Stream}(C)$ is indiscrete for compact-closed C.*

See proof on page 17. As remarked by Bonchi et al. [6, Rem. V.12] this problem can be sidestepped by working in $\mathrm{Stream}_{\mathcal{D}}(C)$ for some non compact-closed symmetric monoidal subcategory $\mathcal{D}$ of C. However this workaround is quite restrictive: in general there is no canonical such subcategory $\mathcal{D}$; moreover, this restriction does not allow the delay to remain a natural transformation.

Similarly in the presence of monoidal zero morphisms, coalgebraic monoidal streams are also degenerate:

Lemma 2.18. *If C is a SMC with monoidal zero morphisms, then $\mathrm{Stream}(C)$ is indiscrete.*

See proof on page 17. One could also try to work around this, as before, by choosing some symmetric monoidal subcategory $\mathcal{D}$ of C which does not itself posses monoidal zero morphisms. This presents two problems. First, as before, this solution is ad-hoc as

[2] In fact they give a more general definition for premonoidal categories; however, we are only interested in monoidal setting in this paper.

there is no such canonical choice of subcategory in general. Secondly, just as is the case for extensional streams in Lemma 2.16, the resulting category of monoidal streams will fail to have zero morphisms, and thus to detect failures in a satisfactory manner.

2.4 Causal and monotone sequences

Another approach to categorical semantics for stateful processes is to represent a stream *concretely* as a family of morphisms in the base category of *increasingly larger* or *increasingly informative* approximations.

2.4.1 Causal sequences. First, consider the following notion due to Raney [52], which constructs stream transducers so that the value of the stream at every time step does not depend on the value of the stream in the future.

Definition 2.19. Let A and B be sets. A function $f : A^{\mathbb{N}} \to B^{\mathbb{N}}$ is **causal** (called *sequential* by Raney) if for all $x, y \in A^{\mathbb{N}}$ and $n \in \mathbb{N}$,

$$\big(\forall i \leq n.\ x_i = y_i\big) \implies \big(\forall i \leq n.\ f(x)_i = f(y)_i\big).$$

We use the following structure to categorify Raney's notion of causal functions in the setting of symmetric monoidal categories:

Definition 2.20. A **discard category** is a SMC where every object X is equipped with a map $\top_X : X \to I$, called the **discard** such that $\top_{X\otimes Y} = \top_X \otimes \top_Y$ and $\top_I = 1_I$. A **discard functor** is a symmetric monoidal functor preserving the discard morphisms. Moreover, a morphism $f : X \to Y$ is **total** in case $\top_X \circ f = \top_Y$. A discard category in which all morphisms are total is called an **affine symmetric monoidal category**.

All of our examples are discard categories, both quantum and classical:

Example 2.21. Any partial Markov category is a discard category; therefore, so are Cartesian restriction categories, Cartesian bicategories, and Markov categories.

Example 2.22. CPM, CPTP and CPTNI are all discard categories with respect to the trace $\mathrm{Tr} : B(\mathcal{H}) \to \mathbb{C}$.

We obtain the following categorification of Raney's causal functions:

Definition 2.23. Let C be a discard category. The affine symmetric monoidal category of **causal sequences** $\mathrm{Caus}_{\mathbb{N}}(C)$ has:

- **Objects**: infinite sequences $X \in \mathrm{ob}(C)^{\mathbb{N}}$;
- **Morphisms**: a morphism $f : X \to Y$ is a causal sequence, i.e. a family of morphisms

  $$\{f_t : X_0 \otimes \cdots \otimes X_t \to Y_0 \otimes \cdots \otimes Y_t\}_{t\in\mathbb{N}}$$

  such that for all $t \in \mathbb{N}$,

  $$(1_{Y_0\otimes\ldots Y_t} \otimes \top_{Y_{t+1}}) \circ f_{t+1} = f_t \otimes \top_{X_{t+1}}; \tag{4}$$

where the rest of the affine symmetric monoidal category structure is defined pointwise.

This notion of causality is well-behaved in categories where every morphism is total. For example, it recovers the causal functions of Katsumata and Sprunger [56, Thm. 14] in Cartesian categories, and controlled stochastic processes in FinStoch [19, Def. 7.1].

However, this is not the correct notion in categories of partial processes. For example, a process f in $\mathrm{Caus}_{\mathbb{N}}(\mathrm{Par})$ such that f_0 is total, must remain total for all $n > 0$. Similarly, for any constant sequence $\{f^{\otimes n}\} \in \mathrm{Caus}_{\mathbb{N}}(C)$ we must have $\top_Y \circ f = \top_X$, i.e. f must be total. Therefore, causal streams do not accomodate failures, which suggests that the equality from Definition 2.23 needs to be relaxed to an *inequality*

$$(1_{Y_0\otimes\ldots Y_t} \otimes \top_{Y_{t+1}}) \circ f_{t+1} \subseteq f_t \otimes \top_{X_{t+1}}.$$

2.4.2 Preorder structure. We formalise the notion of causality in categories of partial and non-determinsitic processes using the following machinery:

Definition 2.24. A **preorder** on a set X is a binary relation $\subseteq$ on X which is reflexive so that for all $x \in X$, $x \subseteq x$; and transitive so that for all $x, y, z \in X$ such that $x \subseteq y$ and $y \subseteq z$ then $x \subseteq z$. When a preorder is moreover antisymmetric, so that $x \subseteq y$ and $y \subseteq x$ implies $x = y$, then it is called a **poset**.

Definition 2.25. A **preorder enrichment** for a SMC C is a a preorder structure $\subseteq$ on every hom-set $C(X, Y)$ such that for all morphisms f, f', g, and g' of the appropriate type:

- if $f \subseteq f'$ and $g \subseteq g'$ then $f \otimes g \subseteq f' \otimes g'$,
- if $f \subseteq f'$ and $g \subseteq g'$ then $f \circ g \subseteq f' \circ g'$.

A **preorder-enriched functor** is a symmetric monoidal functor between preorder-enriched categories which is monotone on homsets. A **preorder-enriched discard category** is a discard category with a preorder enrichment.

We interpret $f \subseteq g$ as denoting that g is an approximation of f, so that f is more informative than g. We interpret the discard map as an uninformative effect (this interpretation will later be justified in Section 4 when impose coherences between the discard and the preorder enrichment).

Partial Markov categories carry a canonical preorder enrichment:

Example 2.26 ([21, § 3]). Given two parallel maps $f, g : X \to Y$ in a partial Markov category C, say that $f \subseteq g$ in case there exists some map $r : Y \otimes X \to I$ such that:

$$f = (Y \otimes r) \circ (\delta_Y \otimes 1_X) \circ (g \otimes 1_X) \circ \delta_X$$

This defines a partial order enrichment on the SMC C called the **conditional preorder** enrichment.

For Cartesian restriction categories, this coincides with the usual restriction poset enrichment [21, Prop. 3.11]. For example, in Par, $f \subseteq g$ in case the domain of definition of f is smaller than that of g and both functions agree on their domain of definition. Similarly for Cartesian bicategories, this order corresponds to the canonical "subobject" poset enrichement [21, Prop. 3.12]. For example, in Rel $S \subseteq T$ in case S is a subset of T. On the other hand for Markov categories and Cartesian categories the conditional preorder is discrete since there is only one effect $r = \top$.

Our quantum examples also carry symmetric monoidal preorder enrichements. First, Selinger [54] introduced a poset-enrichment on completely positive maps:

Definition 2.27. Two parallel completely positive maps $\Phi, \Psi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{K})$ are related by the **Löwner order** $\Phi \preceq \Psi$ in case $\Psi - \Phi$ is completely positive.

Second, Carrette et al. [14, Def. 5] introduced the following order between morphisms in discard categories:

Definition 2.28. Two parallel morphisms $f, g : X \to Y$ in a discard category are related by the **purification order** $f \subseteq g$ in case there exists some $g_0 : X \to S \otimes Y$ and $f_0 : S \to I$ such that

$$f = (f_0 \otimes 1_Y) \circ g_0, \quad \text{and} \quad g = (\top_S \otimes 1_Y) \circ g_0.$$

The comparison between these orders is not made in [14] where only CPTP and CPM are studied. Interestingly, we find that these two notions coincide in CPTNI.

Proposition 2.29. *The Loewner and purification orders*

(1) *are discrete in* CPTP*,*
(2) *coincide in CPTNI, making* CPTNI *a poset-enriched discard category (in fact DCPO enriched),*
(3) *differ in* CPM*, making* CPM *poset-enriched in the Loewner order and preorder-enriched in the purification order.*

See proof on page 18.

2.4.3 Monotone sequences. Finally, with all of these examples in mind, we can relax Definition 2.23 to the preorder-enriched setting:

Definition 2.30. Let C be a preorder-enriched discard category. The discard category of **monotone sequences** $\mathsf{Mon}_{\mathbb{N}}(C)$ has the same objects as $\mathsf{Caus}_{\mathbb{N}}(C)$. A morphism $f : X \to Y$ is a monotone sequence, i.e. a family of morphisms

$$\{f_t : X_0 \otimes \cdots \otimes X_t \to Y_0 \otimes \cdots \otimes Y_t\}_{t \in \mathbb{N}}$$

such that for all $t \in \mathbb{N}$,

$$(1_{Y_0 \otimes \ldots Y_t} \otimes \top_{Y_{t+1}}) \circ f_{t+1} \subseteq f_t \otimes \top_{X_{t+1}}; \tag{5}$$

where the rest of the discard category structure is defined pointwise.

Bonchi et al. show [6, Prop. 6.7] that, in a Markov category with conditionals, their notion of "causal sequences" (which conflicts with the terminology that we use in this paper) are exactly monotone sequences. Carette et al. define a similar construction for discard categories with pseudo-purification, where they impose moreover that every sequence satisfies the very strong assumption of regularity [14, Def. 8].

Monotone sequences accommodate failure: they can contain a component $f_t = 0$ without forcing the whole family $(f_t)_{t \in \mathbb{N}}$ to be the pointwise-zero morphism in $\mathsf{Mon}_{\mathbb{N}}(C)$. However, failures are still not detected at the level of the whole stream: a failure that appears only at a later time step changes only later components in the family.

This motivates imposing an equivalence relation which takes the information theoretic interpretation of preorder enrichment more seriously, identifying two monotone sequences in case they are compatible and neither is more informative than the other.

3 Observational behaviours

In the previous section, we recalled various semantics for stateful processes. Extensional streams are a local semantics, constraints enforced at one time step cannot propagate infinitely far into the future or the past. Coalgebraic streams sit at the opposite extreme, where the fixed point equation imposes a global constraint. Whereas dinatural streams identify few processes, coalgebraic streams identify many. On the other hand, causal streams and monotone streams are very concrete, and they fail to identify processes which ought to behave the same as each other.

In this section we take another approach, introducing the main contribution of this article. We construct a categorical semantics for stateful processes in which the *global behaviour* of a process is determined by the totality of its *local observations*, together with a consistency condition which imposes that observations made in larger contexts can only *refine* those in a smaller context. We identify two processes when they are *observationally indistinguishable*, i.e. when any observation of the one process can be verified by observing the other process, potentially in a larger context.

Another advantage of our approach, unlike existing stream semantics, is that the contexts need not be restricted to discrete intervals of *time*. Therefore, we can interpret processes which are evolving through more general spaces.

Our semantic construction is parametrised by an arbitrary **index set** I and a collection $\mathcal{J} \subseteq \mathcal{P}_f(I)$ of *finite* subsets of I called **contexts**, which define the local finite regions where a global infinite system can be observed. We ask that the set of contexts posses the following property:

Definition 3.1. $\mathcal{J} \subseteq \mathcal{P}_f(I)$ is **upward directed**, in case for any pair of contexts $j, j' \in \mathcal{J}$ there exists some $j'' \supseteq j \cup j'$ in $\mathcal{J}$.

Given two contexts $j \subseteq j'$, we denote the set-theoretic difference by $j' \setminus j$. We assume a total order on I throughout so that the indexed monoidal product $\otimes_{i \in j}$ is well defined for any subset $j \in \mathcal{J}$. We often omit the symmetry morphisms for simplicity and readability, or explicitly use the **unshuffling morphisms**, defined for arbitrary $\mathbf{X}, \mathbf{Y} \in (\mathrm{ob}(C))^I$ and contexts $j \in \mathcal{J}$ as the following isomorphisms:

$$(\beta_{\mathbf{X},\mathbf{Y}})_j : \bigotimes_{k \in j} (X_k \otimes Y_k) \cong \Big(\bigotimes_{k \in j} X_k\Big) \otimes \Big(\bigotimes_{k \in j} Y_k\Big)$$

The morphisms in our semantic category are constructed as monotone nets acting on I-indexed monoidal products:

Definition 3.2 (Monotone net)**.** Let C be a preorder enriched discard category and let $\mathbf{X}, \mathbf{Y} \in (\mathrm{ob}(C))^I$ be a set of objects in C indexed by I. A **monotone net** $f : \mathbf{X} \to \mathbf{Y}$ in C is an assignment to every $j \in \mathcal{J}$, a morphism $f_j : \bigotimes_{i \in j} X_i \to \bigotimes_{i \in j} Y_i$, such that for all $j \subseteq j' \in \mathcal{J}$:

$$(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \subseteq f_j \otimes \top_{X_{j' \setminus j}}$$

where $Y_j \coloneqq \bigotimes_{k \in j} Y_k$ for any $j \in \mathcal{J}$.

A monotone net is **causal** when all the inclusions are equalities.

We use the term "monotone net" as a categorification of the notion of a "net" coined by Kelly, which denotes a function whose domain is a directed set [34].

Note that monotone nets generalise the notion of monotone sequences. A monotone sequence is a monotone net with:

$$I \coloneqq \mathbb{N} \quad \text{and} \quad \mathcal{J} \coloneqq \{[0, n] \mid \forall n \in \mathbb{N}\}.$$

We interpret a context as a finite region in which properties of some global system can be observed (for monotone sequences, this finite region is interpreted as an interval of time). Under this interpretation, the condition of monotonicity imposes that the observations made in any context approximate the observations which can be made in a larger context.

We introduce an equivalence relation between monotone that witnesses when two nets approximate each other.

Definition 3.3. Given two monotone nets $\mathbf{f}, \mathbf{g} : \mathbf{X} \to \mathbf{Y}$, say that $\mathbf{g}$ ***approximates*** $\mathbf{f}$, written $\mathbf{f} \subseteq \mathbf{g}$, if for every $j \in \mathcal{J}$ there exists some $j' \in \mathcal{J}$ with $j \subseteq j'$ such that

$$(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \ \subseteq\ g_j \otimes \top_{X_{j' \setminus j}}.$$

Lemma 3.4. *Approximation is a preorder.*

See proof on page 18.

Definition 3.5. We say that $\mathbf{f}$ and $\mathbf{g}$ are **observationally equivalent**, in case $\mathbf{f}$ and $\mathbf{g}$ approximate each other, i.e. $\mathbf{f} \subseteq \mathbf{g}$ and $\mathbf{g} \subseteq \mathbf{f}$.

We interpret two processes to be observationally equivalent in case their observations never contradict each other; so that every observation of one process can be shown to approximate an observation in the other and vice versa.

This equivalence relation allows us to state the central contruction of the paper:

Definition 3.6. Given an updward directed set of contexts $\mathcal{J} \subseteq \mathcal{P}_f(I)$ and a preorder-enriched discard category C, the poset-enriched discard category of **observational behaviours**, $\mathrm{Obs}_{I,\mathcal{J}}(C)$, has:

- **objects**: families $\mathbf{X} = \{X_i\}_{i \in I}$;
- **morphisms**: observational equivalence classes of monotone nets $[\mathbf{f}] : \mathbf{X} \to \mathbf{Y}$;
- **identities**: $[\{1_{X_j}\}_{j \in \mathcal{J}}]$;
- **post-enrichment**: $[\mathrm{g}] \subseteq [\mathbf{f}]$ if and only if $\mathbf{g} \subseteq \mathbf{f}$;
- **discard category structure:** contextwise.

Theorem 3.7. *The data above makes* $\mathrm{Obs}_{I,\mathcal{J}}(C)$ *a poset-enriched discard category.*

See proof on page 18. The construction of observational behaviours is functorial in the following sense.

Proposition 3.8. *Let* $F : C \to \mathcal{D}$ *be a preorder-enriched discard functor. There is an induced preorder-enriched discard functor*

$$\mathrm{Obs}_{I,\mathcal{J}}(F) : \mathrm{Obs}_{I,\mathcal{J}}(C) \to \mathrm{Obs}_{I,\mathcal{J}}(\mathcal{D}),$$

defined contextwise. Moreover, if F *reflects the preorder then* $\mathrm{Obs}_{I,\mathcal{J}}(F)$ *is faithful. If* F *reflects the preorder and is full, then* $\mathrm{Obs}_{I,\mathcal{J}}(F)$ *is full and faithful.*

See proof on page 19. Recall that the existence of symmetric monoidal zero morphisms makes the category of coalgebraic streams satisfying Equation (3) is codiscrete. On the other hand in extensional streams, a zero morphism occurring at some time step can not propagate infinitely far into the future, meaning that the global process fails to be a monoidal zero morphism. Moreover, for monotone sequences, zero morphisms can only propagate into the future, but not the past. By contrast, behaviours *detect failures* in the following sense:

Proposition 3.9. *Take a preorder enriched discard category* C *with monoidal zero morphisms, and upward-directed* $\mathcal{J} \subseteq \mathcal{P}_f(I)$*. If a monotone net* $\mathbf{f} : \mathbf{X} \to \mathbf{Y}$ *satisfies* $f_{j_0} = 0$ *for some* $j_0 \in \mathcal{J}$*, then* $[\mathbf{f}] = [\{0\}_{j \in \mathcal{J}}]$*; moreover,* $[\{0\}_{j \in \mathcal{J}}]$ *is a zero morphism.*

See proof on page 19.

4 Behavioural semantics of delayed trace

Under which conditions can stateful processes be interpreted as observational behaviours? We approach this question by exhibiting functors from extensional stateful morphisms to categories of behaviours indexed by either the integers or natural numbers.

4.1 Initialised time evolution

The standard interpretation of stateful processes is their discrete time evolution from an initial state. This produces behaviours indexed by the natural numbers.

Definition 4.1. Let $\mathcal{J} \coloneqq \{[0,t] \mid t \in \mathbb{N}\}$ denote the set of finite intervals in $\mathbb{N}$ containing 0. Given a symmetric monoidal preorder enriched category C, denote the category of **initialised behaviours** over C by $\mathrm{Obs}_{\mathbb{N}}(C) \coloneqq \mathrm{Obs}_{\mathbb{N},\mathcal{J}}(C)$.

The delay endofunctor $\Delta : \mathrm{Obs}_{\mathbb{N}}(C) \to \mathrm{Obs}_{\mathbb{N}}(C)$ is given on objects by $\Delta(\mathbf{X}) \coloneqq \{X_{t-1}\}_{t\in\mathbb{N}}$ with $X_{-1} \coloneqq I$, and on morphisms by

$$\Big([\mathbf{f}] : \mathbf{X} \to \mathbf{Y}\Big) \longmapsto \Big([\{1_I \otimes f_{[0,t-1]}\}_{t\in\mathbb{N}}] : \Delta(\mathbf{X}) \to \Delta(\mathbf{Y})\Big)$$

Recall from Section 2, that initialised stateful morphism sequences can be defined as morphisms in $\mathrm{St}([\mathbb{N}, C])$. A sufficient condition on C to interpret these stateful morphism sequences in $\mathrm{Obs}_{\mathbb{N}}(C)$, is the following:

Definition 4.2. We say that a discard category satisfies **lax naturality** if for any $f : X \to Y$ in C, $\top_Y \circ f \subseteq \top_X$

Example 4.3. All our examples satisfy lax naturality:

- In the category of partial functions, morphisms $s : X \to I$ are in one-to-one correspondence with subsets of X and the restricition order is subset inclusion, $\top_X$ is then the largest subset and thus $s \subseteq \top_X$ for any s. The same argument shows that the category of relations $\mathsf{Rel}(C)$ over any regular category also satisfies lax naturality of discard.
- The above examples are instances of the more general fact: in any partial Markov category, the conditional order satisfies lax naturality [21, Prop. 3.5].
- CPTP, CPTNI and CPM equipped with the purification order satisfy lax naturality, as can be easily verified from Definition 2.28. In particular, by Proposition 2.29 CPTNI with the Loewner order satisfies lax naturality.
- CPM with the Loewner order does not satisfy lax naturality. Indeed, take a normalised state in $a \in \mathcal{B}(\mathcal{H})$ and multiply it by a scalar $k > 1$, then $\mathrm{Tr}(ka) = k\,\mathrm{Tr}(a) = k \not\leq 1$.

The following theorem justifies our claim that observational behaviours are a categorical semantics for Mealy machines:

Theorem 4.4. *If* C *is a preorder-enriched discard category satisfying lax-naturality then there is a symmetric monoidal functor* $\mathrm{St}([\mathbb{N}, C], \Delta) \to \mathrm{Obs}_{\mathbb{N}}(C)$.

See proof on page 21.

Note that, in general, $\mathrm{Obs}_{\mathbb{N}}(C)$ will not form a feedback category for monoidal C, although the existence of a functor from $\mathrm{St}([\mathbb{N}, C])$ ensures that dinaturally equivalent stateful morphisms will have the same interpretation. A natural question to ask is whether all initialised behaviours arise in this way? Categorically, this corresponds to showing that the functor $\mathrm{St}([\mathbb{N}, C]) \to \mathrm{Obs}_{\mathbb{N}}(C)$ is full. We show that this holds for all our examples of interest.

Focusing first on the classical examples captured by partial Markov categories with the conditional preorder (see Example 2.26), we start by showing that the functor defined above is full.

Proposition 4.5. *Given a partial Markov category C with the conditional preorder, $U : \mathrm{St}([\mathbb{N}, C], \Delta) \to \mathrm{Obs}_{\mathbb{N}}(C)$ is full.*

See proof on page 22.

Moreover, exploiting the main result of Bonchi et al. [6, Thm 7.19], we can show that coalgebraic streams over partial Markov categories with ranges, with sliding restricted to the subcategory of total morphisms, also map functorially into observational behaviours. The assumption that morphisms have *ranges* is crucial in their proof, we refer to [6, Def 7.1] for the definition, and simply note that this condition holds for Set, Par, Rel, FinStoch, and for substochastic matrices [6, Lem. 7.2]. It is not known to hold in $\mathsf{BorelStoch}_{\leq 1}$ or in more general bicategories of relations.

Proposition 4.6. *Given a partial Markov category C with ranges and the conditional preorder enrichment, let $\mathsf{Tot}(C)$ be the full subcategory of total morphisms, then $\mathrm{Stream}_{\mathsf{Tot}(C)}(C) \simeq \mathrm{Mon}_{\mathbb{N}}(C)$ and it follows that there is a full functor $\mathrm{Stream}_{\mathsf{Tot}(C)}(C) \to \mathrm{Obs}_{\mathbb{N}}(C)$.*

See proof on page 22.

We have thus given two full functors into $\mathrm{Obs}_{\mathbb{N}}(C)$, the first from stateful morphism sequences with unrestricted sliding, the second from coalgebraic streams with sliding restricted to total morphisms. In particular, all previous calculi reviewed in Section 2 are sound and universal for initialised behaviours. Note that these functors do not factor one another since the sliding equation only holds for total morphisms in $\mathrm{Stream}_{\mathsf{Tot}(C)}(C)$. Moreover, none of these two functors is faithful in general. In order to construct a *complete* coalgebraic calculus of initialised behaviours, these two approaches must somehow be combined: allowing finite sliding of arbitrary morphisms (as in $\mathrm{St}([\mathbb{N}, C], \Delta)$), while restricting coinductive rules to only a subclass of morphisms (as in $\mathrm{Stream}_{\mathsf{Tot}(C)}(C)$).

Turning now to the quantum setting, the fullness of the functor $U : \mathrm{St}([\mathbb{N}, C], \Delta) \to \mathrm{Obs}_{\mathbb{N}}(C)$ can this time be shown by exploiting the *Stinespring dilation theorem* for completely positive maps. This has been axiomatised in Carette et al. [14] via the notion of a *pseudo-purifiable* discard category, which comes equipped with the purification order of Definition 2.28. [14, Lem. 6] shows that CPTP and CPM are pseudo-purifiable but the proof is easily shown to apply to CPTNI as well. Following the exact same argument as [14, Prop. 2] we obtain:

Proposition 4.7. *If C is either* CPTP *or* CPTNI *or* CPM*, equipped with the purification order, $U : \mathrm{St}([\mathbb{N}, C], \Delta) \to \mathrm{Obs}_{\mathbb{N}}(C)$ is full.*

In particular, it follows from Proposition 2.29, that for CPTNI with the Loewner order the functor is full.

4.2 Uninitialised time evolution

When working in compact-closed categories, the simplest way to construct a feedback category of stateful processes is to freely add a *delay* natural transformation, see Proposition 2.10. This is convenient because, if we have an axiomatisation of a compact-closed category C in terms of generators and relations, we directly obtain an axiomatisation of $\mathrm{St}(C)$ by adding one generator for delay and imposing that it is natural with respect to all other generators.

However, it is not clear what the time evolution of these processes should be, as there cannot be a functor from $\mathrm{St}(C)$ to initialised streams or behaviours. For example, signal flow graphs add a natural delay transformation to the category of affine relations [8, 9], but their time-domain interpretation has so far not been functorial.

As we proceed to show, stateful morphisms over compact-closed discard categories satisfying lax naturality can be interpreted functorially as *biinfinite* behaviours, indexed by the integers rather than the natural numbers.

Definition 4.8. Let $\mathbb{Z}^2_{\leq} \coloneqq \{[\ell, r] \mid \ell, r \in \mathbb{Z} : \ell \leq r\}$ denote the set of finite closed intervals in $\mathbb{Z}$. Given a symmetric monoidal preorder enriched category C, denote the category of **uninitialised behaviours** over C by $\mathrm{Obs}_{\mathbb{Z}}(C) \coloneqq \mathrm{Obs}_{\mathbb{Z}, \mathbb{Z}^2_{\leq}}(C)$.

The delay endofunctor $\mathtt{Shft} : \mathrm{Obs}_{\mathbb{Z}}(C) \to \mathrm{Obs}_{\mathbb{Z}}(C)$ is given on objects by $\mathtt{Shft}(\mathbf{X}) \coloneqq \{X_{k-1}\}_{k \in \mathbb{Z}}$ and morphisms by:

$$\Big([\mathbf{f}] : \mathbf{X} \to \mathbf{Y}\Big) \longmapsto \Big([\{f_{[\ell-1, r-1]}\}_{[\ell, r] \in \mathbb{Z}^2_{\leq}}] : \mathtt{Shft}(\mathbf{X}) \to \mathtt{Shft}(\mathbf{Y})\Big)$$

A sequence $\mathbf{f} \in \mathrm{Obs}_{\mathbb{Z}}(C)$ is **shift-invariant** if $\mathtt{Shft}(\mathbf{f}) = \mathbf{f}$.

In order to interpret stateful processes as $\mathbb{Z}$-indexed behaviours, we must have a way of initialising the internal state coherently over any interval $[\ell, r]$. When the base category C is compact-closed, there is a natural choice for the initialisation structure given by the transpose of the discard:

$$\bot_X \coloneqq (1_X \otimes \top_{X^*}) \circ \cup_X .$$

Note that lax naturality in the compact-closed setting makes $\top$ and $\bot$ an adjoint pair (in the sense of monoidal bicategories):

Lemma 4.9. *If C is a preorder-enriched compact-closed discard category satisfying lax naturality, then for any object X of C:*

$$\top_X \circ \bot_X \subseteq 1_I \text{ and } 1_X \subseteq \bot_X \circ \top_X$$

See proof on page 22. This observation allows us to show that $\mathrm{Obs}_{\mathbb{Z}}(C)$ is a feedback category:

Theorem 4.10. *If C is a preorder-enriched compact-closed discard category satisfying lax naturality, then there is a monoidal natural transformation*

$$\delta : 1_{\mathrm{Obs}_{\mathbb{Z}}(C)} \Rightarrow \mathtt{Shft}; \quad \delta_{\mathbf{X}} \coloneqq \Big[\big\{\bot_{X_{\ell-1}} \otimes \big(\bigotimes_{k=\ell}^{r-1} 1_{X_k}\big) \otimes \top_{X_r}\big\}_{[\ell, r] \in \mathbb{Z}^2_{\leq}}\Big],$$

making $\mathrm{Obs}_{\mathbb{Z}}(C)$ into a compact-closed feedback category.

See proof on page 22. Using the universal property of the state construction and lax naturality, we obtain a functor witnessing that uninitialised behaviours are a feedback-preserving categorical semantics for Mealy machines in compact-closed categories:

Corollary 4.11. *Assume C is a preorder-enriched compact-closed discard category satisfying lax naturality. Then there are feedback functors:*

$$\mathrm{St}(C) \to \mathrm{St}([\mathbb{Z}, C], \mathtt{Shft}) \to \mathrm{Obs}_{\mathbb{Z}}(C)$$

See proof on page 23.

Example 4.12. Among our examples, these results apply to (i) non-deterministic processes in any Cartesian bicategory $\mathsf{Rel}(C)$ and (ii) unnormalised quantum processes in CPM with the purification order. Indeed both are compact-closed and satisfy lax naturality. We use these results in Section 6, to give a time-domain interpretation of signal flow graphs, seen as relations internal to the category of vector spaces and affine transformations.

5 Compactness theorem for relational nets

When do local approximations completely and uniquely determine a global process? Just as series need not converge, neither do behaviours. However, when the category is sufficiently well-behaved, we may be able to glue the local approximations into a coherent infinite process.

In this section, we show that this is the case for the category of closed relations between compact Hausdorff spaces, exhibiting a functor $\mathsf{Obs}(\mathsf{Rel}(\mathsf{KHaus})) \to \mathsf{Rel}(\mathsf{KHaus})$.

In first-order logic, the *compactness theorem* states that a set of first-order sentences has a model if and only if every finite subset of it has a model (see Gödel [25, Satz X] and later Tarski [57, Thm. 13]). The construction of this functor can be interpreted as a categorified compactness theorem for relational models.

Definition 5.1. Let KHaus denote the category of **compact Hausdorff topological spaces** and **continuous functions**.

Lemma 5.2. *[27, prop 15.]* KHaus *is a regular category, such that the morphisms $X \to Y$ in* $\mathsf{Rel}(\mathsf{KHaus})$ *are exactly the* closed subspaces *of $X \times Y$ with respect to the product topology.*

Hausdorffness is needed for the identity to be closed; whereas compactness is needed for composition to preserve closedness.

We exhibit a functor $\mathsf{Obs}(\mathsf{Rel}(\mathsf{KHaus})) \to \mathsf{Rel}(\mathsf{KHaus})$ which glues together observational equivalence classes of monotone nets on $\{X_j\}_{j\in\mathcal{J}}$ into a closed subspace of $\prod_{j\in\mathcal{J}} X_j$, subject to a covering condition on the set of contexts. We moreover show that this restricts to a *faithful* functor $\mathsf{Obs}(\mathsf{Rel}(\mathsf{FSet})) \to \mathsf{Rel}(\mathsf{KHaus})$.

For an index set I and a set of finite contexts $j, j' \in \mathcal{J}$ such that $j \subseteq j'$, we use the following notations for projections:

$$\pi_j^\infty : \prod_{k\in I} X_k \to \prod_{k\in j} X_k, \quad \text{and} \quad \pi_j^{j'} : \prod_{k\in j'} X_k \to \prod_{k\in j} X_k.$$

Using the inverse image of these projections and intersections, we glue together monotone nets using the following construction:

Lemma 5.3. *Given an I-indexed set of compact Hausdorff spaces $\mathbf{X} = \{X_k \in \mathrm{ob}(\mathsf{KHaus})\}_{k\in I}$, and a monotone net of closed subspaces $\mathbf{C} := \{C_j \subseteq \prod_{k\in j} X_k\}_{j\in\mathcal{J}}$, then there is a compact, closed subspace*

$$\mathtt{lim}(\mathbf{C}) := \bigcap_{j\in\mathcal{J}} (\pi_j^\infty)^{-1}(C_j) \subseteq \prod_{k\in I} X_k.$$

See proof on page 19. The construction of Lemma 5.3 lifts to a functor when the set of contexts posesses the following property:

Definition 5.4. $\mathcal{J} \subseteq \mathcal{P}_f(I)$ is a **cover**, in case for any $i \in I$ there exists a subset $j \in \mathcal{J}$ containing $i \in j$. $\mathcal{J}$ is **cofinal**, in case for each finite subset $s \in \mathcal{P}_f(I)$ there is some $j \supseteq s$ in $\mathcal{J}$. Note that $\mathcal{J}$ is cofinal if and only if it is an updward directed cover.

Theorem 5.5 (Compactness). *Given a set I with cofinal $\mathcal{J} \subseteq \mathcal{P}_f(I)$, then* lim *induces a poset-enriched discard functor:*

$$\mathrm{Lim} : \mathsf{Obs}_{I,\mathcal{J}}(\mathsf{Rel}(\mathsf{KHaus})) \longrightarrow \mathsf{Rel}(\mathsf{KHaus}).$$

See proof on page 19. Note that the assumption that $\mathcal{J}$ is a cover is only used preservation of the identity, while the condition of directedness is only used in for preservation of composition.

Moreover, when each finite context admits only finitely many possible observations, the global behaviour *uniquely determines* which local observations are possible:

Corollary 5.6 (Faithfulness). *Given a set I with cofinal $\mathcal{J} \subseteq \mathcal{P}_f(I)$,* Lim *restricts to a faithful poset-enriched discard functor:*

$$\mathrm{Lim} : \mathsf{Obs}_{I,\mathcal{J}}(\mathsf{Rel}(\mathsf{FSet})) \longrightarrow \mathsf{Rel}(\mathsf{KHaus}).$$

This follows because all descending chains of subsets of finite sets terminate (see Lemma E.2 and Lemma E.3 for a detailed argument). However, because every infinite compact Hausdorff space has an infinite strictly descending chain of subsets, FSet is the largest *full* subcategory of KHaus which admits this faithfulness property.

6 Linear discrete time-invariant systems

Linear time-invariant (LTI) systems are central objects of study in signal processing [47], control theory [58], and electrical circuits [45, § 3.1.3]. They have received increased attention in the context of categorical semantics, thanks to the work of Bonchi, Zanasi et al. [8, 9, 11, 59] as well as Baez et al. [3–5].

While their categorical semantics is well-understood from the perspective of the frequency domain (which we will call the z-domain), particularly with the use of affine relations over fields of polynomials [4, 11], their time-domain interpretation has long eluded categorical treatment. Different approaches exist to capture the "executions" of LTI systems, from the behaviours of Willems [58] and Oberst et al. [45, § 3.1.1] to the trajectories of Bonchi et al. [11], but none is compositional.

In this section, we show that $\mathsf{Obs}_{\mathbb{Z}}$ and $\mathsf{Obs}_{\mathbb{N}}$ give a categorical semantics for uninitialised and initialised LTI systems in the time-domain. We relate the z-domain interpretation of [4, 11] to our time-domain semantics via an oplax feedback functor capturing the inverse z-transform. We moreover show that our notion of uninitialised behaviour, instantiated in this setting, precisely captures the behaviours of Willems [58]. We end by discussing the syntactic implications for the theory of signal flow graphs.

6.1 The z-domain

Fix a field $\mathbb{K}$. Let $\mathsf{FAff}_{\mathbb{K}}$ denote the SMC of affine transformations between finite dimensional-vector spaces, and let $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}$ denote the full subcategory of $\mathsf{Rel}(\mathsf{FAff}_{\mathbb{K}})$ whose objects are finite-dimensional vector spaces. The category of finite-dimensional affine relations, $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}$, is a categorical semantics for time-independent open $\mathbb{K}$-linear systems [4, 7]. The linear relations are interpreted as global constraints; whereas elements of $\mathbb{K}^n$, regarded as affine displacements, are interpreted as signals propagating though the system according to these constraints.

Let $\mathbb{K}[[z]]$ denote the ring of formal power series, and $\mathbb{K}((z))$ its field of fractions, the formal Laurent series (see Section B for a review of the theory of polynomial-type rings). Following Oberst et al. [45, Eq. (8.2) & Rem. 8.1.1], we identify initialised discrete-time signals in the *z-domain*[3] with formal power series $a(z) = \sum_{t=0}^{\infty} a_t z^t$, so that the signal at time t is given by the coeffcient a_t, where multiplication by z corresponds to a discrete time-delay.

By extending scalars along $\mathbb{K}((z)) \supset \mathbb{K}[[z]]$, the delay becomes invertible, so that by forgetting initial conditions, LTI systems in the z-domain may be conveniently represented in terms of $\mathbb{K}((z))$-linear subspaces of the vector space $\mathbb{K}((z))^{\mathsf{in}} \times \mathbb{K}((z))^{\mathsf{out}}$ [45, Cor. & Def. 8.1.4].

Just as linear time-independent systems can be composed in $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}$, following Baez and Erbele [4] as well as Bonchi et al. [9, § 6.2], unitialised LTI systems in the z-domain can be composed in $\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}((z))}$. In order to also account for the signals, we moreover need to include affine displacements, which altogether yields the following feedback category:

Lemma 6.1. *There is faithful symmetric monoidal functor $\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}((z))} \rightarrowtail \mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))}$ and a monoidal natural transformation $\partial : 1_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}((z))}} \Longrightarrow 1_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}((z))}}$; $\partial_V := \{(\mathbf{v}, z \cdot \mathbf{v}) \mid \mathbf{v} \in V\}$, making $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))}$ into a feedback category.*

Note that in order to compose LTI systems in the z-domain using relational composition we are forced to use formal Laurent series, rather than formal power series, since $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}[[z]]} \cong \mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))}$ [59, Thm. 4.14]. Therefore, this relational categorical semantics for LTI systems in the z-domain can not accommodate for initial conditions.

6.2 The time-domain

In the *time-domain*, initialised discrete-time signals are represented by $\mathbb{N}$-indexed sequences in $\mathbb{K}$ [45, § 5.2.2]. Accordingly, the *behaviours* or *trajectories* of initialised LTI systems can be represented by systems of linear constant coefficient *difference equations*, the discrete analogue of differential equations [45, § 3.1.1]. The solutions to these difference equations are certain $\mathbb{K}[z]$-linear subspaces of $(\mathbb{K}^{\mathbb{N}})^{\mathsf{in}} \times (\mathbb{K}^{\mathbb{N}})^{\mathsf{out}}$.

The **inverse z-transform** is the $\mathbb{K}[z]$-linear transformation $\mathcal{Z}_n^\circ : \mathbb{K}[[z]]^n \to (\mathbb{K}^{\mathbb{N}})^n$ which extracts coefficients, transforming signals in the z-domain into the time-domain [45, Rem. 8.1.1].

In order to remain compatible with the categorical semantics for the z-domain, we abandon initial conditions, and define the delay $\Delta : \mathbb{K}^{\mathbb{Z}} \to \mathbb{K}^{\mathbb{Z}}$ by $(\Delta x)_{k+1} := x_k$. In this setting we can now define inverse z-transform as the $\mathbb{K}[z, \frac{1}{z}]$-linear transformation $\mathcal{Z}_n^\circ : \mathbb{K}((z))^n \to (\mathbb{K}^{\mathbb{Z}})^n$ which extracts coefficients.

Note that the inverse z-transform preserves the time-delay:

$$\mathcal{Z}_n^\circ\big(z \cdot f(z)\big) = \Delta\big(\mathcal{Z}_n^\circ(f(z))\big).$$

For each context $[\ell, r] \in \mathbb{Z}^2_{\leq}$ postcomposing this $\mathcal{Z}_n^\circ$ with the projection $(\mathbb{K}^{\mathbb{Z}})^n \to \prod_{k=\ell}^{r} \mathbb{K}^n$, yields the $\mathbb{K}$-linear map $\zeta^n_{[\ell,r]} : \mathbb{K}((z))^n \to \prod_{k=\ell}^{r} \mathbb{K}^n$. This allows us to categorify the inverse z-transform, making $\mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$ a categorical semantics for uninitialised linear time-dependent systems in the time-domain.

Theorem 6.2. *There is an oplax normal feedback functor*

$$\mathcal{Z}^* : \mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))} \to \mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}),$$

given on objects by $n \mapsto n^{\mathbb{Z}}$ and on morphisms $S : n \to m$ by

$$\mathcal{Z}^*(S) \;:=\; \Big[\Big\{\big\{\big(\zeta^n_j(\mathbf{x}), \zeta^m_j(\mathbf{y})\big) \;\big|\; (\mathbf{x}, \mathbf{y}) \in S\big\}\Big\}_{j \in \mathbb{Z}^2_{\leq}}\Big],$$

Where oplax and normal means that

$$\mathcal{Z}^*(T \circ S) \subseteq \mathcal{Z}^*(T) \circ \mathcal{Z}^*(S), \qquad \mathcal{Z}^*(1_n) = 1_{n^{\mathbb{Z}}},$$

See proof on page 23. By replacing $\mathbb{Z}$ with $\mathbb{N}$ we can therefore interpret $\mathsf{Obs}_{\mathbb{N}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$ as a categorical semantics for *initialised* linear time-dependent systems in the time-domain.

Compactness. Given a field $\mathbb{K}$, let $\mathsf{Aff}_{\mathbb{K}}$ denote the SMC of affine transformations between (possibly infinite) vector spaces and the empty set; moreover, let $\mathsf{AR}_{\mathbb{K}}$ denote the full subcategory of $\mathsf{Rel}(\mathsf{Aff}_{\mathbb{K}})$ whose objects are vector spaces. Using Theorem 5.5, and observing that only the *finite* fields are compact, it follows that:

Proposition 6.3. *Given a finite field $\mathbb{F}_q$, and a cofinal set $\mathcal{J} \subseteq \mathcal{P}_f(I)$ there is a faithful poset-enriched discard functor*

$$\mathrm{Lim} : \mathsf{Obs}_{I,\mathcal{J}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{F}_q}) \to \mathsf{AR}_{\mathbb{F}_q}.$$

See proof on page 23. Therefore, observing that $\mathbb{Z}^2_{\leq} \subseteq \mathcal{P}_f(\mathbb{Z})$ and $\{[0, t] \mid t \in \mathbb{N}\} \subseteq \mathcal{P}_f(\mathbb{N})$ are both cofinal it follows that for finite fields $\mathbb{F}_q$, there are faithful poset-enriched discard functors

$$\mathsf{Obs}_{\mathbb{N}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{F}_q}) \to \mathsf{AR}_{\mathbb{F}_q} \quad \text{and} \quad \mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{F}_q}) \to \mathsf{AR}_{\mathbb{F}_q},$$

witnessing a "global" behavioural semantics both for initialised and unitialised linear time-dependent systems. Moreover, the induced oplax functor $\mathrm{Lim} \circ \mathcal{Z}^*$ is given on morphisms by the $\mathbb{F}_q$-linear extension of $\mathcal{Z}_n^\circ : \mathbb{F}_q((z))^n \to (\mathbb{F}_q^{\mathbb{Z}})^n$.

Following Willems, the behaviour of an LTI system is a $\mathbb{K}$-subspace of trajectories cut out by a finite system of constant-coefficient difference equations [58]. Note that the work of Willems, was one of the original motivations of Zanasi for developing $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{F}_q(z)}$ as a categorical semantics for LTI systems [59, P. 12].

The name "behaviour" is no coincidence. Over a finite field, which is the setting where the compactness theorem holds, the induced oplax functor $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{F}_q(z)} \to \mathsf{AR}_{\mathbb{F}_q}$ can be seen to categorify this notion in our setting.

We take the formal definition of behaviours from the book of Oberst et al. [45, § 3.1.1].

Definition 6.4 (Affine behaviour)**.** Let $S \subseteq \mathbb{K}(z)^\ell$ be a finite-dimensional $\mathbb{K}(z)$-linear subspace. Choose $r \in \mathbb{N}$ and a $\mathbb{K}(z)$-linear map $A : \mathbb{K}(z)^\ell \to \mathbb{K}(z)^r$ with $S = \ker A$. Pick $d(z) \in \mathbb{K}[z, z^{-1}] \setminus \{0\}$ clearing denominators and set $B(z) := d(z)A \in \mathbb{K}[z, z^{-1}]^{r \times \ell}$. Evaluate entrywise to obtain the $\mathbb{K}[z, z^{-1}]$-linear map

$$B^{\rho^\ell} : \big((\mathbb{K}^{\mathbb{Z}})^\ell, \rho^\ell\big) \longrightarrow \big((\mathbb{K}^{\mathbb{Z}})^r, \rho^r\big); \qquad B^{\rho^\ell}(u) := B(\Delta)u.$$

Define $\qquad \mathrm{beh}(S) := \ker\big(B^{\rho^\ell}\big) \subseteq (\mathbb{K}^{\mathbb{Z}})^\ell.$

Let $\gamma : \mathbb{K}(z) \to \mathbb{K}((z))$ denote the geometric series expansion and define $\mathcal{Z}^\gamma_\ell(-) := \mathcal{Z}^\circ_\ell(\gamma(-))$. We extend beh to $\mathbb{K}(z)$-affine subspaces $S \subseteq \mathbb{K}(z)^\ell$ by: for $S = \varnothing$ define $\mathrm{beh}(S) := \varnothing$; otherwise choosing some $\mathbf{a} \in S$, with $R := S - \mathbf{a}$: $\mathrm{beh}(S) := \mathrm{beh}(R) + \mathcal{Z}^\gamma_\ell(\mathbf{a}) \subseteq (\mathbb{K}^{\mathbb{Z}})^\ell$.

Note that Oberst et al. [45, § 3.1.1] define this only over linear subspaces, but the extension to affine relations as above is immediately obvious. Indeed the oplax functor $\mathrm{Lim} \circ \mathcal{Z}^*$ can be seen to categorify this notion behaviour, albeit only after restricting to the trajectories which are rational.

[3] Oberst et al. refer to this as the algebraic domain [45, P. 3].

Theorem 6.5. *Let $\mathbb{F}_q$ be a finite field and $S \subseteq \mathbb{F}_q(z)^\ell$ be a finite-dimensional $\mathbb{F}_q(z)$-affine subspace. Then*

$$(\mathrm{Lim} \circ \mathcal{Z}^*)(S) = \mathrm{beh}(S) \cap \mathcal{Z}_\ell^\gamma\big(\mathbb{F}_q(z)^\ell\big).$$

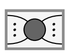
See proof on page 24.

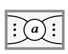
6.3 Signal flow graphs

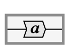
Linear *signal flow graphs* are a syntax for linear time-invariant systems; see for example, the seminal work Shannon [55], and later of Mason [39, 40]. In this section, we discuss how the z-domain and time-domain semantics for unitialised LTI systems produce different equational theories for signal flow graphs; followed briefly by a discussion concerning the syntactic implications of our initialised time-domain semantics.

First we recall the following equational theory for linear *time-independent* systems:

Theorem 6.6 (Bonchi et al. [7]). *Given a field $\mathbb{K}$, the SMC $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}$ is presented by the following generators, for all $a \in \mathbb{K}$:*

$\llbracket$  $\rrbracket := \{(\mathbf{v}, \mathbf{w}) \mid \forall j, k : v_j = w_k\}$,

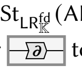
$\llbracket$  $\rrbracket := \{(\mathbf{v}, \mathbf{w}) \mid a + \Sigma\, v_j = \Sigma\, w_j\}$,

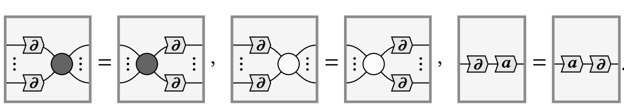
$\llbracket$  $\rrbracket := \{(b, ab) \mid \forall b \in \mathbb{K}\}$

modulo the equations given in Appendix A,

These generators impose linear constraints on systems; where the label a on the white note is interpreted as a signal propagating through this linear system.

The question of finding an equational theory for LTI systems, and axiomatizing signal flow graphs, can therefore be reduced to the question of how to add a notion of *time* to the equational theory of $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}$.

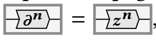
The simplest way to answer this question is to freely add a discrete-time delay generator which commutes only with the constraints, which by Proposition 2.10 is captured by the feedback category $\mathsf{St}_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$. Concretely, this is presented by adding the generator  to the equational theory of $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}$ modulo the equations:

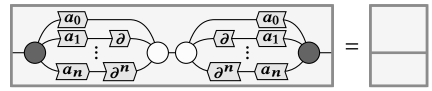


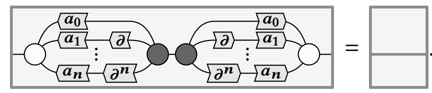
Where the white spider with no label denotes a white spider with label 0 (i.e. with no affine shift).

Let $\mathbb{K}(z)$ denote the field of fractions of the polynomial ring $\mathbb{K}[z]$ (i.e. the rational functions over $\mathbb{K}$). Just as $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))}$ is a feedback category, so is $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}(z)}$. Moreover,

Lemma 6.7. *There is a full feedback functor $\mathsf{St}_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}) \twoheadrightarrow \mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}(z)}$, given by identifying the delays and extending scalars along $\mathbb{K}(z)/\mathbb{K}$, and extending scalars along the geometric series embedding $\mathbb{K}((z))/\mathbb{K}(z)$ induces a feedback functor $\mathtt{Ext} : \mathsf{St}_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}) \to \mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))}$.*

See proof on page 24. Identifying the respective delay maps, so that , the equational theory of $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}(z)}$ is given by quotienting the equational theory of $\mathsf{St}_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$ by the following equations for all nonzero $\mathbf{a} \in \mathbb{K}[z]$:

 (6)

 (7)

Where the left facing arrows denote the transposes of the right facing arrows.

We can also interpret the "free" syntax for LTI systems in our categorical semantics for the time-domain so that the affine translations are interpreted as impulses at time-step 0 and the linear subspaces are invariant with respect to time:

Lemma 6.8. *Let $\eta : \mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}} \to \mathsf{St}_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$ be the canonical functor. Define $J : \mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}} \to \mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$ by $J(V)_t := V$ for all $t \in \mathbb{Z}$, and on morphisms by $J(\varnothing) := [\{\varnothing\}_{j \in \mathbb{Z}^2_{\leq}}]$ and*

$$J(A + \mathbf{a}) := \left[\left\{ \bigoplus_{t=\ell}^{r} \begin{matrix} A + \mathbf{a} & \textit{if } t = 0, \\ A & \textit{if } t \neq 0 \end{matrix} \right\}_{[\ell, r] \in \mathbb{Z}^2_{\leq}}\right].$$

Then there is a unique feedback functor

$$\mathtt{Unroll} : \mathsf{St}_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}) \to \mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}) \textit{ satisfying } \mathtt{Unroll} \circ \eta = J.$$

See proof on page 24. It is natural to ask if invertibility equations (6) and (7) also hold in $\mathtt{Unroll}\,\mathsf{St}_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$; this is only partly true:

Proposition 6.9. *With respect to polynomials $z + a \in \mathbb{K}[z]$ for $a \neq 0$, $\mathcal{Z}^*(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))})$ satisfies Equation (7) but only satisfies Equation (6) oplaxly.*

See proof on page 25. In particular, this means, at least that for algebraically closed $\mathbb{K}$, all nonzero polynomials in $\mathbb{K}[z]$ have left inverse in $\mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$. However, only polynomials of the form az^n for nonzero $a \in \mathbb{K}$ and $n \in \mathbb{N}$ have right inverses, where for all other nonzero polynomials the right inverse equation holds oplaxly. Nevertheless, in a *relaxed* sense, both semantics are compatible:

Proposition 6.10. *The following diagram laxly commutes:*

$$\begin{tikzcd}
\mathsf{St}_{\mathsf{LR}^{\mathsf{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}) \arrow[r, "\mathtt{Unroll}"] \arrow[d, two heads, "\mathtt{Ext}"'] \arrow[dr, phantom, "\subseteq"] & \mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}) \\
\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))} \arrow[ur, bend right, "\mathcal{Z}^*"'] &
\end{tikzcd}.$$

That is to say, $\mathcal{Z}^*(\mathtt{Ext}(-)) \subseteq \mathtt{Unroll}(-)$.

See proof on page 25. The obstruction to invertibility is a consequence that the solutions to difference equations are determined only up to initial conditions. This can be resolved by working in $\mathsf{Obs}_{\mathbb{N}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}})$ and initialising the relevant monotone sequences to begin with the zero subspace. Developing the syntax of initialised systems leaves much to be discovered.

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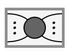
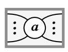
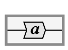
A Graphical affine algebra

The following result is due to Bonchi et al. [7]; however, for convenience, we use the "spider" notation, which is given in [12]:

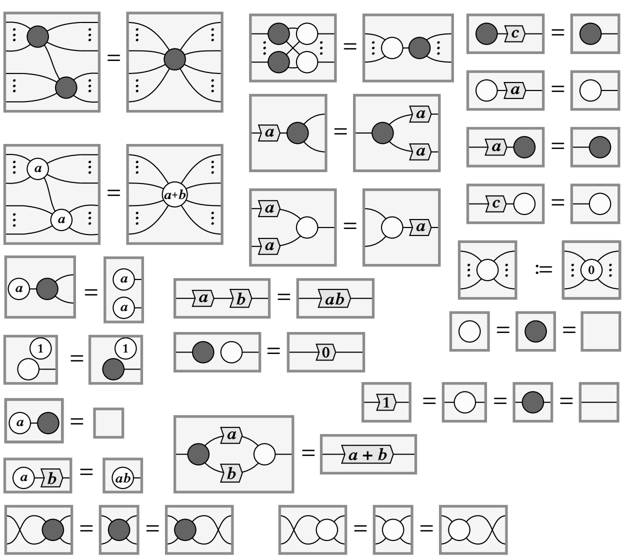
Theorem 6.6. *Given a field $\mathbb{K}$, the SMC $\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}$ is presented by the generators, for all $a \in \mathbb{K}$:*

$[\![$  $]\!] \coloneqq \{(\mathbf{v}, \mathbf{w}) \mid \forall j, k : v_j = w_k\}$,

$[\![$  $]\!] \coloneqq \{(\mathbf{v}, \mathbf{w}) \mid a + \Sigma\, v_j = \Sigma\, w_j\}$,

$[\![$  $]\!] \coloneqq \{(b, ab) \mid \forall b \in \mathbb{K}\}$,

modulo the following equations, for all $a, b \in \mathbb{K}$ and $c \in \mathbb{K}^\times$:



B Rings of polynomials

In this Appendix we recall some basic algebraic notions involving rings of polynomials. In order to establish notation, we start with the following definition which we all know and love:

Definition B.1. Given a field $\mathbb{K}$, the **polynomial ring** $\mathbb{K}[z]$ is a principal ideal domain (PID).

This can be extended to a larger PID by localising z

Definition B.2. The PID $\mathbb{K}[z, 1/z]$ given by adding an inverse to the polynomial indeterminate z in $\mathbb{K}[z]$ is called the PID of **Laurent polynomials** over $\mathbb{K}$ in z.

Lemma B.3. *There is an embedding of PIDs*

$$\mathbb{K}[z] \rightarrowtail \mathbb{K}[z, 1/z].$$

The polynomial ring can also be extended to a larger PID:

Definition B.4. The PID of **formal power series** $\mathbb{K}[[z]]$ has elements given by infinite sequences in $\mathbb{K}$, denoted

$$f(z) \coloneqq \sum_{j=0}^{N} f_j \cdot z^j$$

The addititive, Abelian group structure is given pointwise as for polynomials:

$$f(z) + g(z) \coloneqq \sum_{j=0}^{\infty} (f_j + g_j) \cdot z^j,$$

$$-f(z) \coloneqq \sum_{j=0}^{\infty} (-f_j) \cdot z^j, \quad \text{and} \quad 0 \coloneqq 0 \cdot z^1.$$

The multiplicative monoid structure is given by:

$$f(z)g(z) \coloneqq \sum_{j=0}^{\infty} \sum_{k=-\infty}^{\infty} (f_j \cdot g_k) \cdot z^{j+k}, \text{ and } 1 \coloneqq 1 \cdot z^0$$

Lemma B.5. *There is an embedding of PIDs*

$$\mathbb{K}[z] \rightarrowtail \mathbb{K}[[z]].$$

The polynomial ring can be extended to a field:

Definition B.6. The field of fractions of polynomials or Laurent polynomials in z, $\mathbb{K}(z)$ is called the field of **rational functions** over $\mathbb{K}$ in z.

Lemma B.7. *There is an embedding of PIDs*

$$\mathbb{K}[z, 1/z] \rightarrowtail \mathbb{K}(z).$$

By taking the z-adic completion, this can be extended to an even bigger field, which is a bit more involved to define:

Definition B.8. The field of **formal Laurent series** $\mathbb{K}((z))$ is the z adic completion of $\mathbb{K}[z]$, $\mathbb{K}[z, 1/z]$, $\mathbb{K}[[z]]$ or $\mathbb{K}(z)$, which is more complicated to describe concretely. $\mathbb{K}((z))$ has elements which are biinfinite sequences $f(z) \in \mathbb{K}^{\mathbb{Z}}$ of elements in $\mathbb{K}$, bounded below some $m \in \mathbb{Z}$, denoted:

$$f(z) \coloneqq \sum_{j=m}^{\infty} f_j \cdot z^j$$

The largest j such that for all $k \in \mathbb{N}$ $f_{j-k-1} = 0$, if it exists, is called the **order** of $f(z)$, denoted $\mathrm{ord}(f(z))$; otherwise, set $\mathrm{ord}(f(z)) \coloneqq 0$.

The additive structure group and the multiplicative monoidal structure are essentially the same as for formal power series.

The multiplicative inverse is a bit more complicated. Take some nonzero $f(z)$. Then we can factor out the lowest order term:

$$f(z) = f_{\mathrm{ord}(f(z))} z^{\mathrm{ord}(f(z))} \left(1 + \sum_{j=1}^{\infty} \frac{f_{j+\mathrm{ord}(f(z))}}{f_{\mathrm{ord}(f(z))}} z^j\right)$$

The inverse of $1 + \sum_{j=1}^{\infty} \frac{f_{j+\mathrm{ord}(f(z))}}{f_{\mathrm{ord}(f(z))}} z^j$ is therefore a geometric series $g(z) \coloneqq \sum_{j=0}^{\infty} g_j \cdot z^j$ where the coefficients are given by the recursive formula:

$$g_0 = 1, \quad \text{where} \quad g_{k+1} = -\sum_{k=1}^{j+1} \frac{f_{k+\mathrm{ord}(f)}}{f_{\mathrm{ord}(f)}} g_{j-k+1}$$

This gives us the inverse $f^{-1}(z) \coloneqq \frac{z^{-\operatorname{ord}(f(z))}}{f_{\operatorname{ord}(f(z))}} g(z)$.

Lemma B.9. *There is a field extension*

$$\gamma : \mathbb{K}(z) \rightarrowtail \mathbb{K}((z)),$$

*called the **geometric series embedding**, sending $f(z)/g(z) \in \mathbb{K}(z)$ to $f(z) \cdot g^{-1}(z) \in \mathbb{K}((z))$.*

Putting this all together we have the lattice of PIDs:

$$\begin{tikzcd}
\mathbb{K} \arrow[r, tail] & \mathbb{K}[z] \arrow[r, tail] \arrow[d, tail] & \mathbb{K}[z,1/z] \arrow[r, tail] & \mathbb{K}(z) \arrow[d, tail] \\
& \mathbb{K}[[z]] \arrow[rr, tail] & & \mathbb{K}((z))
\end{tikzcd}$$

C Proofs of Section 2

Lemma C.1. *Take two symmetric monoidal functors $\Gamma : \mathcal{D} \to \mathcal{C}$ and $\Delta : \mathcal{D} \to \mathcal{D}$, and let $\eta : \mathcal{C} \to \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ be the canonical symmetric monoidal functor. Then $\mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ carries a canonical feedback structure given by $\Delta : \mathcal{D} \to \mathcal{D}$,*

$$\widehat{\Gamma} \coloneqq \eta \circ \Gamma : \mathcal{D} \to \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta), \quad \textit{and} \quad \mathsf{fbk}_S([M, f]) \coloneqq [M \otimes S, f].$$

Proof. Fix $S \in \mathrm{ob}(\mathcal{D})$ and $X, Y \in \mathrm{ob}(\mathcal{C})$. Since η is the identity on objects,

$$\widehat{\Gamma}(S) = \Gamma(S), \qquad \widehat{\Gamma}(\Delta S) = \Gamma(\Delta S).$$

Take a representative

$$[M, f] \in \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)\big(\widehat{\Gamma}(\Delta S) \otimes X,\ \widehat{\Gamma}(S) \otimes Y\big).$$

Thus $M \in \mathrm{ob}(\mathcal{D})$ and

$$f \in \mathcal{C}\big(\Gamma(\Delta M) \otimes \Gamma(\Delta S) \otimes X,\ \Gamma(M) \otimes \Gamma(S) \otimes Y\big).$$

Define $\mathsf{fbk}_S([M, f]) \coloneqq [M \otimes S, f]$. This is well-typed since Δ and Γ are strict symmetric monoidal, hence

$$\Gamma(\Delta(M \otimes S)) = \Gamma(\Delta M) \otimes \Gamma(\Delta S), \qquad \Gamma(M \otimes S) = \Gamma(M) \otimes \Gamma(S),$$

so f is also an element of

$$\mathcal{C}\big(\Gamma(\Delta(M \otimes S)) \otimes X,\ \Gamma(M \otimes S) \otimes Y\big),$$

and thus, $[M \otimes S, f] \in \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(X, Y)$.

Independence of representatives. Fix $t \in \mathcal{D}(M, M')$ and

$$f \in \mathcal{C}\big(\Gamma(\Delta M) \otimes \Gamma(\Delta S) \otimes X,\ \Gamma(M') \otimes \Gamma(S) \otimes Y\big).$$

Write $A \coloneqq \widehat{\Gamma}(\Delta S) \otimes X$ and $B \coloneqq \widehat{\Gamma}(S) \otimes Y$. Then

The feedback operation in $\mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ satisfies the sliding equation on representatives:

$$\begin{aligned}
&\mathsf{fbk}_{M'}(M' \otimes S, (\widehat{\Gamma}(t) \otimes 1_B) \circ f) \\
&\quad = (S, (\widehat{\Gamma}(t) \otimes 1_B) \circ f) \\
&\quad = (S, f \circ (\widehat{\Gamma}(\Delta t) \otimes 1_A)) \\
&\quad = \mathsf{fbk}_M(M \otimes S, f \circ (\widehat{\Gamma}(\Delta t) \otimes 1_A)).
\end{aligned}$$

Therefore fbk_M is well-defined.

The other feedback category axioms follow from the evident diagram chases. □

This is the *free* feedback category, in the following sense:

Proposition C.2. *Take two symmetric monoidal functors $\Gamma : \mathcal{D} \to \mathcal{C}$ and $\Delta : \mathcal{D} \to \mathcal{D}$. Given a feedback category $(\mathcal{C}', \mathcal{D}', \Gamma', \Delta', \mathsf{fbk}')$ and two symmetric monoidal functors $F : \mathcal{C} \to \mathcal{C}'$ and $F|_{\mathcal{D}} : \mathcal{D} \to \mathcal{D}'$ satisfying $\Gamma' \circ F|_{\mathcal{D}} = F \circ \Gamma$ and $\Delta' \circ F|_{\mathcal{D}} = F|_{\mathcal{D}} \circ \Delta$, there is a unique feedback functor $(F^\sharp, F|_{\mathcal{D}}) : \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta) \to \mathcal{C}'$ making the diagram commute:*

$$\begin{tikzcd}
\mathcal{C} \arrow[r, "\eta"] \arrow[dr, "F"'] & \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta) \arrow[d, dotted, "{(F^\sharp, F|_{\mathcal{D}})}"] \\
& \mathcal{C}'.
\end{tikzcd}$$

Proof. Define $F^\sharp : \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta) \to \mathcal{C}'$ on objects by $F^\sharp(X) \coloneqq F(X)$. Given some $[S, f] \in \mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)(X, Y)$, define

$$F^\sharp([S, f]) \coloneqq \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\big(F(f)\big) \in \mathcal{C}'(F(X), F(Y)).$$

This is well-typed since

$$\Gamma' \circ F|_{\mathcal{D}} = F \circ \Gamma, \quad \text{and} \quad \Delta' \circ F|_{\mathcal{D}} = F|_{\mathcal{D}} \circ \Delta;$$

therefore,

$$F(\Gamma(S)) = \Gamma'(F|_{\mathcal{D}}(S)), \quad \text{and} \quad F(\Gamma(\Delta S)) = \Gamma'(\Delta'(F|_{\mathcal{D}}(S))).$$

Well-definedness. Fix

$$s \in \mathcal{D}(S, S'), \quad \text{and} \quad f \in \mathcal{C}(\Gamma(\Delta S') \otimes X,\ \Gamma(S) \otimes Y).$$

In $\mathrm{St}_{\mathcal{D}}(\mathcal{C}, \Delta)$ we have

$$[S', (\Gamma(s) \otimes 1_Y) \circ f] = [S, f \circ (\Gamma(\Delta s) \otimes 1_X)].$$

On the other hand in $\mathcal{C}'$, we have

$$\begin{aligned}
&\mathsf{fbk}'_{F|_{\mathcal{D}}(S')}\Big(F\big((\Gamma(s) \otimes 1_Y) \circ f\big)\Big) \\
&\quad = \mathsf{fbk}'_{F|_{\mathcal{D}}(S')}\Big((F(\Gamma(s)) \otimes 1_{F(Y)}) \circ F(f)\Big) \\
&\quad = \mathsf{fbk}'_{F|_{\mathcal{D}}(S')}\Big((\Gamma'(F|_{\mathcal{D}}(s)) \otimes 1_{F(Y)}) \circ F(f)\Big) \\
&\quad = \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\Big(F(f) \circ (\Gamma'(\Delta'(F|_{\mathcal{D}}(s))) \otimes 1_{F(X)})\Big) \\
&\quad = \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\Big(F(f) \circ (\Gamma'(F|_{\mathcal{D}}(\Delta s)) \otimes 1_{F(X)})\Big) \\
&\quad = \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\Big(F(f) \circ (F(\Gamma(\Delta s)) \otimes 1_{F(X)})\Big) \\
&\quad = \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\Big(F(f) \circ F(\Gamma(\Delta s) \otimes 1_X)\Big) \\
&\quad = \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\Big(F\big(f \circ (\Gamma(\Delta s) \otimes 1_X)\big)\Big)
\end{aligned}$$

Hence $F^\sharp$ is well-defined.

Functoriality and symmetric monoidality. follow from the evident diagram chases.

Triangle commutes. For $f \in \mathcal{C}(X, Y)$,

$$\begin{aligned}
F^\sharp(\eta(f)) = F^\sharp([I, f]) &= \mathsf{fbk}'_{F|_{\mathcal{D}}(I)}(F(f)) \\
&= \mathsf{fbk}'_I(F(f)) = F(f),
\end{aligned}$$

Compatibility with Γ and Δ. By assumption

$$\Delta' \circ F|_{\mathcal{D}} = F|_{\mathcal{D}} \circ \Delta;$$

moreover,

$$F^\sharp \circ \widehat{\Gamma} = F^\sharp \circ \eta \circ \Gamma = F \circ \Gamma = \Gamma' \circ F|_{\mathcal{D}}.$$

Preservation of feedback. Let $S \in \mathrm{ob}(\mathcal{D})$ and take a representative

$$[M,h] \in \mathrm{St}_{\mathcal{D}}(\mathcal{C},\Delta)\big(\widehat{\Gamma}(\Delta S)\otimes X,\ \widehat{\Gamma}(S)\otimes Y\big).$$

Then

$$\begin{aligned} F^\sharp(\mathsf{fbk}_S([M,h])) = F^\sharp([S\otimes M,h]) &= \mathsf{fbk}'_{F|_{\mathcal{D}}(S\otimes M)}(F(h)) \\ &= \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\Big(\mathsf{fbk}'_{F|_{\mathcal{D}}(M)}(F(h))\Big) \\ &= \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\big(F^\sharp([M,h])\big), \end{aligned}$$

Uniqueness. Take another feedback functor

$$(H, F|_{\mathcal{D}}) : \mathrm{St}_{\mathcal{D}}(\mathcal{C},\Delta) \to \mathcal{C}',$$

such that $H \circ \eta = F$. Moreover take any representative

$$(S,f) \in \mathrm{St}_{\mathcal{D}}(\mathcal{C},\Delta)(\widehat{\Gamma}(\Delta S)\otimes X, (\widehat{\Gamma}S)\otimes Y.$$

Therefore

$$\begin{aligned} H([S,f]) &= H(\mathsf{fbk}_S([I,f])) \\ &= H(\mathsf{fbk}_S(\eta f)) = \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\big(H(\eta f)\big) \\ &= \mathsf{fbk}'_{F|_{\mathcal{D}}(S)}\big(F(f)\big) = F^\sharp([S,f]). \end{aligned}$$

Hence $H = F^\sharp$, so $(F^\sharp, F|_{\mathcal{D}})$ is unique. □

Proposition 2.10. *Take a compact-closed category $\mathcal{C}$, a faithful symmetric monoidal functor $\Gamma : \mathcal{D} \rightarrowtail \mathcal{C}$ such that for all $X \in \mathrm{ob}(\mathcal{D})$, $\Gamma^{-1}(\delta_X) \neq \varnothing$, and a symmetric monoidal endofunctor $\Delta : \mathcal{D} \to \mathcal{D}$. The existence of a feedback operator is equivalent to a monoidal natural transformation $\delta : 1_{\mathcal{D}} \Rightarrow \Delta$, called the* ***delay natural transformation****. Moreover, such a natural transformation is necessarily a natural isomorphism.*

Proof of Proposition 2.10. First we prove the biconditional; after which we prove that the delay natural transformation is necessarily an isomorphism.

Assume that for all objects $S \in \mathcal{D}$, we have a feedback operator:

$$\mathsf{fbk}_S : \mathcal{C}(\Gamma(\Delta S)\otimes X, \Gamma(S)\otimes Y) \to \mathcal{C}(X,Y).$$

Using the fact that $\Gamma : \mathcal{D} \rightarrowtail \mathcal{C}$ is a faithful symmetric monoidal functor, for all $S \in \mathrm{ob}(\mathcal{D}$, define $\delta_S \in \mathcal{D}(S, \Delta S)$ by:

$$\delta_S \coloneqq \Gamma^{-1}\Big(\mathsf{fbk}_S\big(\mathsf{swap}_{\Gamma(\Delta S),\Gamma S}\big)\Big) \in \mathcal{C}(\Gamma(S), \Gamma(\Delta S)).$$

which exists by assumption and is well-defined due to the faithfulness of Γ.

To prove naturality, define

$$f \coloneqq \mathsf{swap}_{\Gamma(\Delta S),\Gamma(S)} \in \mathcal{C}(\Gamma(\Delta T)\otimes\Gamma(S),\ \Gamma(S)\otimes\Gamma(\Delta T)).$$

Take any $h \in \mathcal{D}(S,T)$. Using sliding, we have

$$\mathsf{fbk}_T\big((\Gamma(h)\otimes 1_{\Gamma(\Delta T)})\circ f\big) = \mathsf{fbk}_S\big(f\circ(\Gamma(\Delta h)\otimes 1_{\Gamma(S)})\big).$$

Using the naturality of swap, we have

$$(\Gamma(h)\otimes 1)\circ \mathsf{swap}_{\Gamma(\Delta T),\Gamma(S)} = \mathsf{swap}_{\Gamma(\Delta T),\Gamma(T)} \circ (1\otimes\Gamma(h)),$$

and similarly

$$\mathsf{swap}_{\Gamma(\Delta T),\Gamma(S)} \circ (\Gamma(\Delta h)\otimes 1) = (1\otimes \Gamma(\Delta h))\circ \mathsf{swap}_{\Gamma(\Delta S),\Gamma(S)}.$$

Substituting these into the sliding equation gives:

$$\begin{aligned} &\mathsf{fbk}_T\big(\mathsf{swap}_{\Gamma(\Delta T),\Gamma(T)} \circ (1\otimes\Gamma(h))\big) \\ &\quad = \mathsf{fbk}_S\big((1\otimes\Gamma(\Delta h))\circ \mathsf{swap}_{\Gamma(\Delta S),\Gamma(S)}\big). \end{aligned}$$

Now applying tightening twice yields

$$\begin{aligned} &\mathsf{fbk}_T\big(\mathsf{swap}_{\Gamma(\Delta T),\Gamma(T)} \circ (1\otimes\Gamma(h))\big) \\ &\quad = \mathsf{fbk}_T\big(\mathsf{swap}_{\Gamma(\Delta T),\Gamma(T)}\big)\circ\Gamma(h) \\ &\quad = \Gamma(\delta_T)\circ\Gamma(h), \end{aligned}$$

and similarly

$$\begin{aligned} &\mathsf{fbk}_S\big((1\otimes\Gamma(\Delta h))\circ \mathsf{swap}_{\Gamma(\Delta S),\Gamma(S)}\big) \\ &\quad = \Gamma(\Delta h)\circ \mathsf{fbk}_S\big(\mathsf{swap}_{\Gamma(\Delta S),\Gamma(S)}\big) \\ &\quad = \Gamma(\Delta h)\circ\Gamma(\delta_S). \end{aligned}$$

Hence

$$\Gamma(\Delta h)\circ\Gamma(\delta_S) = \Gamma(\delta_T)\circ\Gamma(h).$$

Since Γ is faithful, this implies that $\Delta h\circ\delta_S = \delta_T\circ h$, therefore $\delta : 1_{\mathcal{D}} \Rightarrow \Delta$ is natural. Monoidality of δ follows from strength, joining, and vanishing together with coherence of swap.

Conversely, assume we have a monoidal natural transformation $\delta : 1_{\mathcal{D}} \Rightarrow \Delta$. Take $S \in \mathrm{ob}(\mathcal{D})$ and $f \in \mathcal{C}(\Gamma(\Delta S)\otimes X, \Gamma(S)\otimes Y)$. Define

$$\begin{aligned} \mathsf{fbk}_S(f) \coloneqq\ & (\cap_{\Gamma(S)}\otimes 1_Y)\circ(\mathsf{swap}_{\Gamma(S),\Gamma(S)^*}\otimes 1_Y) \\ &\circ (1_{\Gamma(S)}\otimes \mathsf{swap}_{Y,\Gamma(S)^*}) \\ &\circ \big((f\circ(\Gamma(\delta_S)\otimes 1_X))\otimes 1_{\Gamma(S)^*}\big) \\ &\circ (1_{\Gamma(S)}\otimes \mathsf{swap}_{\Gamma(S)^*,X})\circ(\cup_{\Gamma(S)}\otimes 1_X). \end{aligned}$$

The feedback axioms then follow from compact closure together with the fact that δ is a monoidal natural transformation, using essentially the same argument as is spelled out in [36].

Now we prove that the delay natural transformation is necessarily an isomorphism.

Consider the delay natural transformation $\delta : 1_{\mathcal{D}} \Rightarrow \Delta$ from Proposition 2.10. Define $\gamma : \Delta \Rightarrow 1_{\mathcal{D}}$ componentwise by

$$\gamma_X \coloneqq (\cap_{\Delta(X^*)}\otimes 1_X)\circ(1_{\Delta X}\otimes\delta_{X^*}\otimes 1_X)\circ(1_{\Delta X}\otimes\cup_{X^*}).$$

Naturality of γ follows by applying the dual $(-)^*$ to the naturality square for δ.

We show that γ is inverse to δ. Because Δ is a symmetric monoidal functor on a compact-closed category for all objects D in $\mathcal{D}$ we have:

$$\cup_{\Delta Z} = \Delta(\cup_Z) \quad\text{and}\quad \cap_{\Delta Z} = \Delta(\cap_Z)$$

For the one direction, naturality of δ at $\cap_{X^*} : X\otimes X^* \to I$, together with monoidality of δ, gives

$$\cap_{\Delta(X^*)}\circ(\delta_X\otimes\delta_{X^*}) = \cap_{X^*}.$$

Hence

$$\begin{aligned} \gamma_X\circ\delta_X &= (\cap_{\Delta(X^*)}\otimes 1_X)\circ(1_{\Delta X}\otimes\delta_{X^*}\otimes 1_X) \\ &\qquad \circ(1_{\Delta X}\otimes\cup_{X^*})\circ\delta_X \\ &= (\cap_{\Delta(X^*)}\otimes 1_X)\circ(\delta_X\otimes\delta_{X^*}\otimes 1_X)\circ(1_X\otimes\cup_{X^*}) \\ &= (\cap_{X^*}\otimes 1_X)\circ(1_X\otimes\cup_{X^*}) = 1_X. \end{aligned}$$

For the other direction, naturality of δ at $\cup_{X^*} : I \to X^*\otimes X$, together with monoidality of δ, gives

$$(\delta_{X^*}\otimes\delta_X)\circ\cup_{X^*} = \cup_{\Delta(X^*)}.$$

Therefore

$$\begin{aligned}\delta_X \circ \gamma_X &= \delta_X \circ (\cap_{\Delta(X^*)} \otimes 1_X) \circ (1_{\Delta X} \otimes \delta_{X^*} \otimes 1_X) \\ &\quad \circ (1_{\Delta X} \otimes \cup_{X^*}) \\ &= (\cap_{\Delta(X^*)} \otimes 1_{\Delta X}) \circ (1_{\Delta X} \otimes \delta_{X^*} \otimes \delta_X) \\ &\quad \circ (1_{\Delta X} \otimes \cup_{X^*}) \\ &= (\cap_{\Delta(X^*)} \otimes 1_{\Delta X}) \circ (1_{\Delta X} \otimes \cup_{\Delta(X^*)}) = 1_{\Delta X}.\end{aligned}$$

Thus γ is a two-sided inverse to δ, so δ is a natural isomorphism. □

Lemma 2.12. *For every SMC C, there is a unique feedback functor $U : \mathrm{St}(C) \to \mathrm{St}([\mathbb{Z}, C], \mathtt{Shft})$ sending:*

$$\Big([S, f] : X \to Y\Big) \;\mapsto\; \Big(\big[(f : S \otimes X \to S \otimes Y)_{j\in\mathbb{Z}}\big] : X^{\mathbb{Z}} \to Y^{\mathbb{Z}}\Big).$$

Proof of Lemma 2.12. Define the symmetric monoidal functor

$$J : C \to \mathrm{St}([\mathbb{Z}, C], \mathtt{Shft}); \quad (f : X \to Y) \mapsto [(I \otimes f)_{j\in\mathbb{Z}}]$$

Using the canonical functor $\eta : C \to \mathrm{St}(C)$, by the universal property of the state construction given in Proposition C.2, there exists a unique feedback functor

$$J^\sharp : \mathrm{St}(C) \to \mathrm{St}([\mathbb{Z}, C], \mathtt{Shft}),$$

such that for all $f \in C(X, Y)$

$$J^\sharp([I, f]) = [(f)_{j\in\mathbb{Z}}].$$

Moreover, for any $[S, f] \in \mathrm{St}(C)(X, Y)$, $J^\sharp$ being a feedback functor means that

$$J^\sharp([S, F]) = \mathtt{Shft}(J^\sharp([S, F])),$$

which yields the desired claim. □

To further understand coalgebraic streams, we must further expose the work of Di Lavore et al. [19].

Definition C.3 ([19, Def. 4.2])**.** Fix a symmetric monoidal category $(C, \otimes, I)$ and $\mathbf{X}, \mathbf{Y} \in \mathrm{ob}(C)^{\mathbb{N}}$. Define $M_{-1} \coloneqq I$ and for all $n \in \mathbb{N}$ define

$$\mathrm{Stage}_n(\mathbf{X}, \mathbf{Y}) \coloneqq \int^{\mathrm{M}\in C^{n-1}} \prod_{i=0}^{n} C(M_{i-1} \otimes X_i,\ M_i \otimes Y_i),$$

denoting representatives by $[\ \langle f_0| \cdots |f_n|\ ] \in \mathrm{Stage}_n(\mathbf{X}, \mathbf{Y})$. Moreover, define the truncation maps by:

$$\begin{aligned}\pi_n : \mathrm{Stage}_{n+1}(\mathbf{X}, \mathbf{Y}) &\to \mathrm{Stage}_n(\mathbf{X}, \mathbf{Y}); \\ \langle f_0| \cdots |f_{n+1}| &\mapsto \langle f_0| \cdots |f_n|.\end{aligned}$$

Definition C.4 ([19, Def. 9])**.** A SMC is **productive** if for all $\alpha \in \mathrm{Stage}_0(\mathbf{X}, \mathbf{Y})$ there exists a representative $\alpha_0 : X_0 \to M_0 \otimes Y_0$ such that, for every representative $\alpha' : X_0 \to M \otimes Y_0$ of α, there exists $s : M_0 \to M$ with $\alpha' = (s \otimes 1_{Y_0}) \circ \alpha_0$.

Lemma C.5. *The following SMCs are productive:*

- *Cartesian monoidal categories;*
- *compact-closed categories;*
- *SMCs with monoidal zero morphisms.*

Proof. The first two claims are proved by Di Lavore et al. [19, Remark 4.10]. Therefore, we only prove the final claim involving monoidal zero morphisms.

Fix $\mathbf{X}, \mathbf{Y} \in \mathrm{ob}(C)^{\mathbb{N}}$ and let $\alpha \in \mathrm{Stage}_0(\mathbf{X}, \mathbf{Y})$. Choose a representative $\alpha = \langle \alpha_0|$ with $\alpha_0 : X_0 \to M \otimes Y_0$ for some $M \in \mathrm{ob}(C)$. By the axioms for monoidal zero morphisms

$$0_{M,I} \otimes 1_{Y_0} = \mathsf{swap}_{I,Y_0} \circ (1_{Y_0} \otimes 0_{M,I}) \circ \mathsf{swap}_{M,Y_0} = 0_{M\otimes Y_0,\, I\otimes Y_0}.$$

By dinaturality

$$\langle \alpha_0| = \langle (0_{M,I} \otimes 1_{Y_0}) \circ \alpha_0| = \langle 0_{X_0,\, I\otimes Y_0}|.$$

Thus $\mathrm{Stage}_0(\mathbf{X}, \mathbf{Y})$ is a singleton. Hence every 0-stage process is terminating, and C is productive. □

Lemma C.6 ([19, Thm. 4.11.]). *If C is productive, then for all $\mathbf{X}, \mathbf{Y} \in \mathrm{ob}(C)^{\mathbb{N}}$ there is canonical a bijection* $\mathrm{Stream}(C)(\mathbf{X}, \mathbf{Y}) \cong \lim_{n\in\mathbb{N}} \mathrm{Stage}_n(\mathbf{X}, \mathbf{Y})$*, where the limit is taken along the truncations π_n.*

Lemma 2.17. $\mathrm{Stream}(C)$ *is indiscrete for compact-closed C.*

Proof of Lemma 2.17. Fix $\mathbf{X}, \mathbf{Y} \in \mathrm{ob}(C)^{\mathbb{N}}$. Take some

$$\langle f_0|f_1| \cdots |f_n| \in \mathrm{Stage}_n(\mathbf{X}, \mathbf{Y})$$

with witnesses $f_i : M_{i-1} \otimes X_i \to M_i \otimes Y_i$ and $M_{-1} \coloneqq I$. Define

$$\eta_{M,X,Y} \coloneqq 1_{M\otimes X} \otimes (\mathsf{swap}_{Y,Y^*} \circ \cup_Y) : M \otimes X \to (M \otimes X \otimes Y^*) \otimes Y,$$

and for any $f : M \otimes X \to N \otimes Y$ set

$$s_f \coloneqq (1_N \otimes \cap_Y) \circ (1_N \otimes \mathsf{swap}_{Y,Y^*}) \circ (f \otimes 1_{Y^*}) : M \otimes X \otimes Y^* \to N,$$

so $(s_f \otimes 1_Y) \circ \eta_{M,X,Y} = f$. Write $N_{-1} \coloneqq I$ and $N_k \coloneqq N_{k-1} \otimes X_k \otimes Y_k^*$. Then by dinaturality of the coends and the above factorisation,

$$\begin{aligned}&\langle f_0| \cdots |f_{n-1}|f_n| \\ &\approx \langle f_0| \cdots |f_{n-1}|\eta_{M_{n-1},X_n,Y_n}| \\ &\approx \langle f_0| \cdots |f_{n-2}|\eta_{M_{n-2},X_{n-1},Y_{n-1}}|\eta_{M_{n-2}\otimes X_{n-1}\otimes Y_{n-1}^*,X_n,Y_n}| \\ &\ \ \vdots \\ &= \langle \eta_{N_{-1},X_0,Y_0}|\eta_{N_0,X_1,Y_1}| \cdots |\eta_{N_{n-1},X_n,Y_n}|.\end{aligned}$$

Therefore for all $n \in \mathbb{N}$, $\mathrm{Stage}_n(\mathbf{X}, \mathbf{Y})$ is a singleton. By Lemma C.5, C is productive, so Lemma C.6 implies $\mathrm{Stream}(C)$ is indiscrete. □

Lemma 2.18. *If C is a SMC with monoidal zero morphisms, then* $\mathrm{Stream}(C)$ *is indiscrete.*

Proof of Lemma 2.18. Fix $\mathbf{X}, \mathbf{Y} \in \mathrm{ob}(C)^{\mathbb{N}}$. Take some

$$\langle f_0|f_1| \cdots |f_n| \in \mathrm{Stage}_n(\mathbf{X}, \mathbf{Y})$$

with witnesses

$$f_n : M_{n-1} \otimes X_n \to M_n \otimes Y_n.$$

Then by dinaturality and the coherences for monoidal zero morphisms

$$\begin{aligned}\langle f_0| \cdots |f_{n-1}|f_n| &\approx \langle f_0| \cdots |f_{n-1}|(0_{M_n,I} \otimes 1_{Y_n}) \circ f_n| \\ &= \langle f_0| \cdots |f_{n-1}|0_{M_n\otimes X_n, I\otimes Y_n}| \\ &\ \ \vdots \\ &= \langle 0_{X_0,Y_0}| \cdots |0_{X_{n-1},Y_{n-1}}|0_{X_n,Y_n}|\end{aligned}$$

Therefore for all $n \in \mathbb{N}$, $\mathrm{Stage}_n(\mathbf{X}, \mathbf{Y})$ is a singleton. By Lemma C.5, C is productive, so Lemma C.6 $\mathrm{Stream}(C)$ is indiscrete. □

Proposition 2.29. *The Loewner and purification orders*

(1) *are discrete in* CPTP*,*
(2) *coincide in CPTNI, making* CPTNI *a poset-enriched discard category (in fact DCPO enriched),*
(3) *differ in* CPM*, making* CPM *poset-enriched in the Loewner order and preorder-enriched in the purification order.*

Proof of Proposition 2.29. (1) The purification order in CPTP is discrete since the only effect is the trace operator. To see that the Loewner order is discrete, suppose $\Phi \preceq \Psi$, the $\Psi - \Phi$ is completely positive, then for any normalised state $a \in \mathcal{B}(\mathcal{H})$, $\Psi(a) - \Phi(a)$ must be positive, moreover since these maps preserve the trace $\mathrm{Tr}(\Psi(a)) = \mathrm{Tr}(\Phi(a)) = \mathrm{Tr}(a) = 1$, but then $\mathrm{Tr}(\Psi(a) - \Phi(a)) = 0$ by linearity of the trace. The only CP map that sends all normalised states to states of trace 0 is the zero map, therefore $\Psi = \Phi$.

(2) Selinger shows [54, Lem. 6.4] is a poset enrichement for CPTNI, DCPO-enrichment is proved by Kenta Cho [16, Thm. 5.5]. It remains to show that the two orders coincide. First, suppose that $\Phi \preceq \Psi$, then $\Xi = \Psi - \Phi$ is completely positive. Consider the qubit space $\mathbb{C}^2$ with basis $|0\rangle, |1\rangle$ and define $g_0 : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{K} \otimes \mathbb{C}^2)$ and $f_0 : \mathcal{B}(\mathbb{C}^2) \to \mathbb{C}$ by:

$$g_0 = \Phi \otimes |0\rangle\langle 0| + \Xi \otimes |1\rangle\langle 1| \qquad f_0(a) = \langle 0| a |0\rangle$$

for any $a \in \mathcal{B}(\mathbb{C}^2)$. It is easy to check that these maps are trace non-increasing and that $\Phi = (1 \otimes f_0) \circ g_0$ and $\Psi = (1 \otimes \mathrm{Tr}) \circ g_0$. Therefore $\Phi \subseteq \Psi$ in the purification order. Now for the other direction, suppose $\Phi \subseteq \Psi$ then there exists trace non-increasing maps g_0, f_0 such that $\Phi = (1 \otimes f_0) \circ g_0$ and $\Psi = (1 \otimes \mathrm{Tr}) \circ g_0$. Then we have $\Psi - \Phi = (1 \otimes (\mathrm{Tr} - f_0)) \circ g_0$ and since f_0 is trace non-increasing $\mathrm{Tr} - f_0$ is completely positive. Since the composition of completely positive maps is completely positive we obtain $\Psi \subseteq \Phi$.

(3) CPM is preorder-enriched in the purification order by [14, Lem. 6]. A similar argument to [54, Lem. 6.4] shows that CPM is poset-enriched in the Loewner order. To see that these two orders do not coincide we can simply look at scalars. Any positive real number is a scalar in CPM. For any pair of scalars $a, b \in \mathbb{R}^+$, taking $g_0 = b$ and $f_0 = \frac{a}{b}$ we have that $a \subseteq b$ in the purification order. However if $b \leq a$ the inequality does not hold in the Loewner order. □

D Proofs of Section 3

Lemma 3.4. *Approximation is a preorder.*

Proof of Lemma 3.4. Reflexivity is immediate by taking $j' = j$. For transitivity, suppose $\mathbf{f} \subseteq \mathbf{g}$ and $\mathbf{g} \subseteq \mathbf{h}$. Fix $j \in \mathcal{J}$. By $\mathbf{g} \subseteq \mathbf{h}$, there exists $j_0 \in \mathcal{J}$ with $j \subseteq j_0$ such that

$$(1_{Y_j} \otimes \top_{Y_{j_0 \setminus j}}) \circ g_{j_0} \subseteq h_j \otimes \top_{X_{j_0 \setminus j}}.$$

By $\mathbf{f} \subseteq \mathbf{g}$, applied to the context j_0, there exists $j_1 \in \mathcal{J}$ with $j_0 \subseteq j_1$ such that

$$(1_{Y_{j_0}} \otimes \top_{Y_{j_1 \setminus j_0}}) \circ f_{j_1} \subseteq g_{j_0} \otimes \top_{X_{j_1 \setminus j_0}}.$$

Composing and reassociating discards gives

$$(1_{Y_j} \otimes \top_{Y_{j_1 \setminus j}}) \circ f_{j_1} \subseteq h_j \otimes \top_{X_{j_1 \setminus j}},$$

so $\mathbf{f} \subseteq \mathbf{h}$. □

Theorem 3.7. *The data above makes* $\mathrm{Obs}_{I,\mathcal{J}}(C)$ *a poset-enriched discard category.*

Proof of Theorem 3.7. We need to show that:

(1) every structural morphism is represented by a monotone net;
(2) composition and monoidal product send monotone nets to monotone nets;
(3) the preorder on monotone nets descends to observational equivalence classes and satisfies the axioms of a preorder enrichment;
(4) identities, monoidal unit, symmetry, discard, composition, and tensor product satisfy the axioms of discard categories.

(1) Structural morphisms are monotone. Identities, symmetry, and discard are defined contextwise by tensoring the corresponding structural morphisms in C:

$$(1_{\mathbf{X}})_j = 1_{X_j}, \qquad (\top_{\mathbf{X}})_j = \bigotimes_{i \in j} \top_{X_i},$$

and similarly for symmetry.

Let $j \subseteq j'$. Because $\top \circ f = \top$ for all structural morphisms f in C, we have

$$(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} = f_j \otimes \top_{X_{j' \setminus j}},$$

so the monotonicity inequalities hold as equalities. Hence all structural morphisms define causal (in particular, monotone) nets.

(2) Composition and tensor preserve monotonicity. Let $\mathbf{f} : \mathbf{X} \to \mathbf{Y}$ and $\mathbf{g} : \mathbf{Y} \to \mathbf{Z}$ be composable monotone nets. Fix $j \subseteq j'$ in $\mathcal{J}$. Using monotonicity of $\mathbf{g}$ and functoriality of composition, we compute:

$$\begin{aligned}(1_{Z_j} \otimes \top_{Z_{j' \setminus j}}) \circ (g_{j'} \circ f_{j'}) &= \big((1_{Z_j} \otimes \top_{Z_{j' \setminus j}}) \circ g_{j'}\big) \circ f_{j'} \\ &\subseteq \big(g_j \otimes \top_{Y_{j' \setminus j}}\big) \circ f_{j'} \\ &\subseteq (g_j \circ f_j) \otimes \top_{X_{j' \setminus j}},\end{aligned}$$

where the last inclusion uses monotonicity of $\mathbf{f}$. Thus $\mathbf{g} \circ \mathbf{f}$ is monotone.

Now let $\mathbf{f} : \mathbf{X} \to \mathbf{Y}$ and $\mathbf{h} : \mathbf{X}' \to \mathbf{Y}'$ be monotone nets. For $j \subseteq j'$, we compute:

$$\begin{aligned}&(1_{(Y \otimes Y')_j} \otimes \top_{(Y \otimes Y')_{j' \setminus j}}) \circ (\beta_{\mathrm{Y},\mathrm{Y}'})_j^{-1} \circ (f_{j'} \otimes h_{j'}) \circ (\beta_{\mathrm{X},\mathrm{X}'})_j \\ &= (\beta_{\mathrm{Y},\mathrm{Y}'})_j^{-1} \circ (1_{Y_j} \otimes \top_{Y_{j' \setminus j}} \otimes 1_{Y'_j} \otimes \top_{Y'_{j' \setminus j}}) \\ &\qquad \circ (f_{j'} \otimes h_{j'}) \circ (\beta_{\mathrm{X},\mathrm{X}'})_j \\ &\subseteq (\beta_{\mathrm{Y},\mathrm{Y}'})_j^{-1} \circ (f_j \otimes h_j) \circ (\beta_{\mathrm{X},\mathrm{X}'})_j.\end{aligned}$$

Hence $\mathbf{f} \otimes \mathbf{h}$ is monotone.

(3) Compatibility with the preorder. Since observational equivalence is the symmetric closure of approximation, it suffices to show that composition and tensor are monotone with respect to $\subseteq$.

Suppose $\mathbf{f} \subseteq \mathbf{f}'$ and $\mathbf{g} \subseteq \mathbf{g}'$, and that the pairs are composable. Fix $j \in \mathcal{J}$. Choose $j_f, j_g \in \mathcal{J}$ with $j \subseteq j_f$ and $j \subseteq j_g$ witnessing that $\mathbf{f}'$ approximates $\mathbf{f}$ and $\mathbf{g}'$ approximates $\mathbf{g}$. Using upward directedness of $\mathcal{J}$, let $j' \in \mathcal{J}$ be such that $j_f \cup j_g \subseteq j'$.

Then:

$$\begin{aligned}(1_{Z_j} \otimes \top_{Z_{j' \setminus j}}) \circ g_{j'} \circ f_{j'} &\subseteq g'_j \circ (1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ f_{j'} \\ &\subseteq (g'_j \circ f'_j) \otimes \top_{X_{j' \setminus j}}.\end{aligned}$$

Thus $\mathbf{g} \circ \mathbf{f} \subseteq \mathbf{g}' \circ \mathbf{f}'$. An analogous argument shows that tensor product is monotone with respect to $\subseteq$.

Therefore composition and tensor are well defined on equivalence classes.

(4) Discard category axioms. All discard-category axioms (functoriality of discard, compatibility with tensor, symmetry, unit, and composition) hold contextwise by the corresponding axioms in C. Since observational equivalence respects contextwise equality, the axioms lift to $\mathrm{Obs}_{I,\mathcal{J}}(C)$.

This completes the proof. □

Proposition 3.8. *Let $F : C \to \mathcal{D}$ be a preorder-enriched discard functor. There is an induced preorder-enriched discard functor*

$$\mathrm{Obs}_{I,\mathcal{J}}(F) : \mathrm{Obs}_{I,\mathcal{J}}(C) \to \mathrm{Obs}_{I,\mathcal{J}}(\mathcal{D}),$$

defined contextwise. Moreover, if F reflects the preorder then $\mathrm{Obs}_{I,\mathcal{J}}(F)$ is faithful. If F reflects the preorder and is full, then $\mathrm{Obs}_{I,\mathcal{J}}(F)$ is full and faithful.

Proof of Proposition 3.8. We proceed in several steps.

Preservation of monotonicity. Fix $j \subseteq j'$ in $\mathcal{J}$. By monotonicity of $\mathbf{f}$,

$$(1_{Y_j} \otimes \top_{Y_{j'\setminus j}}) \circ f_{j'} \subseteq f_j \otimes \top_{X_{j'\setminus j}}.$$

Applying F and using that it preserves the preorder gives

$$F\big((1_{Y_j} \otimes \top_{Y_{j'\setminus j}}) \circ f_{j'}\big) \subseteq F\big(f_j \otimes \top_{X_{j'\setminus j}}\big).$$

Since F is a discard functor and symmetric monoidal, this rewrites as

$$(1_{F(Y)_j} \otimes \top_{F(Y)_{j'\setminus j}}) \circ F(f_{j'}) \subseteq F(f_j) \otimes \top_{F(X)_{j'\setminus j}},$$

which is exactly the monotonicity condition for $\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{f})$. Hence $\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{f})$ is a monotone net in $\mathcal{D}$.

Well-definedness on equivalence classes. Suppose $\mathbf{f} \subseteq \mathbf{g}$ in $\mathrm{Obs}_{I,\mathcal{J}}(C)$. Fix $j \in \mathcal{J}$. By definition, there exists $j' \in \mathcal{J}$ with $j \subseteq j'$ such that

$$(1_{Y_j} \otimes \top_{Y_{j'\setminus j}}) \circ f_{j'} \subseteq g_j \otimes \top_{X_{j'\setminus j}}.$$

Applying F and rewriting as before yields

$$(1_{F(Y)_j} \otimes \top_{F(Y)_{j'\setminus j}}) \circ F(f_{j'}) \subseteq F(g_j) \otimes \top_{F(X)_{j'\setminus j}}.$$

Thus $\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{f}) \subseteq \mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{g})$, so $\mathrm{Obs}_{I,\mathcal{J}}(F)$ is well defined on observational equivalence classes.

Functoriality. Functoriality holds contextwise. For identities, for each $j \in \mathcal{J}$,

$$(\mathrm{Obs}_{I,\mathcal{J}}(F)(1_{\mathbf{X}}))_j = F(1_{X_j}) = 1_{F(X_j)} = (1_{\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{X})})_j.$$

For composition, let $\mathbf{f} : \mathbf{X} \to \mathbf{Y}$ and $\mathbf{g} : \mathbf{Y} \to \mathbf{Z}$. Then for each $j \in \mathcal{J}$,

$$\begin{aligned}(\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{g} \circ \mathbf{f}))_j &= F(g_j \circ f_j) = F(g_j) \circ F(f_j)\\ &= (\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{g}) \circ \mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{f}))_j.\end{aligned}$$

Hence $\mathrm{Obs}_{I,\mathcal{J}}(F)$ is a functor.

Preservation of the symmetric monoidal structure and discard. All structure is preserved contextwise. For example, for tensor products,

$$(\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{f} \otimes \mathbf{h}))_j = F\big((\beta_{\mathbf{Y},\mathbf{Y}'})_j^{-1} \circ (f_j \otimes h_j) \circ (\beta_{\mathbf{X},\mathbf{X}'})_j\big),$$

which equals the corresponding tensor expression built from $F(f_j)$ and $F(h_j)$ since F is symmetric monoidal. Discard preservation follows from

$$F\Big(\bigotimes_{i\in j} \top_{X_i}\Big) = \bigotimes_{i\in j} \top_{F(X_i)}.$$

Thus $\mathrm{Obs}_{I,\mathcal{J}}(F)$ is a preorder-enriched discard functor.

Faithfulness. Assume F reflects the preorder. Suppose that

$$\mathrm{Obs}_{I,\mathcal{J}}(F)([\mathbf{f}]) = \mathrm{Obs}_{I,\mathcal{J}}(F)([\mathbf{g}]).$$

Then $\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{f})$ and $\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{g})$ approximate each other. Fix $j \in \mathcal{J}$. There exists $j' \supseteq j$ such that

$$(1_{F(Y)_j} \otimes \top_{F(Y)_{j'\setminus j}}) \circ F(f_{j'}) \subseteq F(g_j) \otimes \top_{F(X)_{j'\setminus j}}.$$

Rewrite both sides as $F(u) \subseteq F(v)$. By reflection of the preorder, $u \subseteq v$, hence $\mathbf{f} \subseteq \mathbf{g}$. The reverse inclusion is analogous, so $[\mathbf{f}] = [\mathbf{g}]$. Thus $\mathrm{Obs}_{I,\mathcal{J}}(F)$ is faithful.

Fullness. Assume F reflects the preorder and is full. Let $[\mathbf{h}]$: $\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{X}) \to \mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{Y})$ be a morphism in $\mathrm{Obs}_{I,\mathcal{J}}(\mathcal{D})$, represented by a monotone net $\mathbf{h}$. For each $j \in \mathcal{J}$, use fullness to choose f_j in C such that $F(f_j) = h_j$. Let $\mathbf{f} = \{f_j\}_{j\in\mathcal{J}}$.

Monotonicity of $\mathbf{h}$ and the reflection of the preorder imply monotonicity of $\mathbf{f}$. Contextwise, $\mathrm{Obs}_{I,\mathcal{J}}(F)(\mathbf{f}) = \mathbf{h}$, hence $\mathrm{Obs}_{I,\mathcal{J}}(F)$ is full. □

Proposition 3.9. *Take a preorder enriched discard category C with monoidal zero morphisms, and upward-directed $\mathcal{J} \subseteq \mathcal{P}_f(I)$. If a monotone net $\mathbf{f} : \mathbf{X} \to \mathbf{Y}$ satisfies $f_{j_0} = 0$ for some $j_0 \in \mathcal{J}$, then $[\mathbf{f}] = [\{0\}_{j\in\mathcal{J}}]$; moreover, $[\{0\}_{j\in\mathcal{J}}]$ is a zero morphism.*

Proof of Proposition 3.9. Fix $j \in \mathcal{J}$. Choose any $j' \in \mathcal{J}$ with $j \cup j_0 \subseteq j'$. By monotonicity, $f_{j'}$ factors through $f_{j_0} = 0$, hence

$$(1_{Y_j} \otimes \top_{Y_{j'\setminus j}}) \circ f_{j'} = 0.$$

This witnesses $\mathbf{f} \subseteq \{0\}$, and the converse approximation is immediate.

Zero axioms lift componentwise. □

E Proofs of Section 5

Lemma 5.3. *Given an I-indexed set of compact Hausdorff spaces $\mathbf{X} = \{X_k \in \mathrm{ob}(\mathsf{KHaus})\}_{k\in I}$, and a monotone net of closed subspaces $\mathbf{C} := \{C_j \subseteq \prod_{k\in j} X_k\}_{j\in\mathcal{J}}$, then there is a compact, closed subspace*

$$\mathtt{lim}(\mathbf{C}) := \bigcap_{j\in\mathcal{J}} (\pi_j^\infty)^{-1}(C_j) \subseteq \prod_{k\in I} X_k.$$

Proof of Lemma 5.3. The product $\prod_{k\in I} X_k$ is compact with the product topology by Tychonoff's theorem and any intersection of closed subsets of a compact space is closed and compact. □

Theorem 5.5 (Compactness). *Given a set I with cofinal $\mathcal{J} \subseteq \mathcal{P}_f(I)$, then $\mathtt{lim}$ induces a poset-enriched discard functor:*

$$\mathrm{Lim} : \mathrm{Obs}_{I,\mathcal{J}}(\mathsf{Rel}(\mathsf{KHaus})) \longrightarrow \mathsf{Rel}(\mathsf{KHaus}).$$

Proof of Theorem 5.5. Define Lim on objects by

$$\mathbf{X} = \{X_k \in \mathrm{ob}(\mathsf{KHaus})\}_{k\in I} \mapsto \prod_{k\in I} X_k,$$

and on monotone nets $\mathbf{S} : \mathbf{X} \to \mathbf{Y}$ by

$$\mathrm{Lim}(\mathbf{S}) = \Theta_{\mathbf{X},\mathbf{Y}}\big(\mathtt{lim}(\mathbf{S})\big) \subseteq \Big(\prod_{i\in\mathbb{Z}} X_i\Big) \times \Big(\prod_{i\in\mathbb{Z}} Y_i\Big).$$

where $\Theta_{\mathbf{X},\mathbf{Y}} : \prod_{k\in I} X_k \otimes Y_k \cong \prod_{k\in I} X_k \times \prod_{k\in I} Y_k$.

To show this is a valid functor we need it to preserve identities, discards, monoidal unit, the poset structure on monotone nets, composition, and monoidal product.

Preservation of monoidal unit and discards. Lim strictly preserves the monoidal unit since $\prod_{k\in I}\{*\} \cong \{*\}$. Then observe that $\mathtt{lim}(\top_{\mathbf{X}}) = \bigcap_{j\in\mathcal{J}}(\pi_j^\infty)^{-1}(X_j) = \bigcap_{j\in\mathcal{J}}\prod_{k\in I}X_k = \prod_{k\in I}X_k$. Therefore Lim preserves discards.

Preservation of identities. For preservation of identities we want to show $1_{\prod_{k\in I}X_k} = \mathrm{Lim}(1_{\mathbf{X}})$. First, for any j we have $\pi_j^\infty 1_{\prod_{k\in I}X_k} \subseteq 1_{X_j}$ and therefore

$$1_{\prod_{k\in I}X_k} \subseteq \bigcap_{j\in\mathcal{J}}(\pi_j^\infty)^{-1}(1_{X_j}) = \mathrm{Lim}(1_{\mathbf{X}})\,.$$

For the other direction, suppose $(x,y)\in\mathrm{Lim}(1_{\mathbf{X}})$, then $\pi_j^\infty(x,y)\in 1_{X_j}$ and so, for all j, $\pi_j^\infty x = \pi_j^\infty y$. Using that $\mathcal{J}$ is a *cover*, for any $k\in I$ there is $j\ni k$ and projecting out the k^{th} component of X_j we get $x_k = y_k$. Therefore $x=y$ and we have shown $\mathrm{Lim}(1_{\mathbf{X}}) \subseteq 1_{\prod_{i\in I}X_i}$.

Preservation of poset and well-definedness on equivalence classes. Suppose $\mathbf{A}\subseteq\mathbf{B}$, then there is a function f that to any context j associates a larger context $j' = f(j)\supseteq j$ such that $\pi_j^{j'}A_{j'}\subseteq B_j$. Then since $\pi_j^\infty = \pi_j^{j'}\circ\pi_{j'}^\infty$ we have $(\pi_j^\infty)^{-1} = (\pi_{j'}^\infty)^{-1}\circ(\pi_j^{j'})^{-1}$ and so $(\pi_{j'}^\infty)^{-1}(A_{j'}) \subseteq (\pi_j^\infty)^{-1}(B_j)$. Then we have:

$$\begin{aligned}\mathtt{lim}(\mathbf{A}) &= \bigcap_{j\in\mathcal{J}}(\pi_j^\infty)^{-1}(A_j)\\ &\subseteq \bigcap_{j\in\mathcal{J}}(\pi_{f(j)}^\infty)^{-1}(A_{f(j)})\\ &\subseteq \bigcap_{j\in\mathcal{J}}(\pi_j^\infty)^{-1}(B_j) = \mathtt{lim}(\mathbf{B})\end{aligned}$$

Therefore the Lim preserves the poset on monotone nets. It follows that Lim is well-defined on observational equivalence classes.

Preservation of monoidal product. Let $\mathbf{S}$ and $\mathbf{T}$ be two observational nets. Without loss of generality assume $S_j\subseteq A_j$ and $S_j\subseteq B_j$ for some nets $\mathbf{A},\mathbf{B}\in ob(C)^I$. Then $\pi_{j'}^\infty(a,b) = (\pi_{j'}^{\infty,A}a, \pi_{j'}^{\infty,B}b)$ for the projections $\pi_{j'}^{\infty,X}:\prod_{k\in I}X_k\to\prod_{k\in j}X_k$. Therefore:

$$\begin{aligned}\mathtt{lim}(\mathbf{S}\otimes\mathbf{T}) &= \bigcap_{j\in\mathcal{J}}\left((\pi_{j'}^{\infty,A})^{-1}(S_j)\times(\pi_{j'}^{\infty,B})^{-1}(T_j)\right)\\ &= \bigcap_{j\in\mathcal{J}}\left((\pi_{j'}^{\infty,A})^{-1}(S_j)\right)\times\bigcap_{j\in\mathcal{J}}\left((\pi_{j'}^{\infty,B})^{-1}(T_j)\right)\\ &\simeq \mathtt{lim}(\mathbf{S})\times\mathtt{lim}(\mathbf{T})\end{aligned}$$

and it follows that Lim preserves monoidal products.

Preservation of composition: laxness. Let $\mathbf{S}$ and $\mathbf{T}$ be composable nets. We start by showing laxness $\mathrm{Lim}(\mathbf{T})\circ\mathrm{Lim}(\mathbf{S})\subseteq\mathrm{Lim}(\mathbf{T}\circ\mathbf{S})$. Suppose that $(x,z)\in\mathrm{Lim}(\mathbf{T})\circ\mathrm{Lim}(\mathbf{S})$ then there exists y such that $(x,y)\in\mathrm{Lim}(\mathbf{S})$ and $(y,z)\in\mathrm{Lim}(\mathbf{S})$. Note first that for any $j_0\in\mathcal{J}$ $\pi_{j_0}^\infty(\mathbf{C})\subseteq C_{j_0}$ since:

$$\begin{aligned}\pi_{j_0}^\infty(\mathtt{lim}(\mathbf{C})) &= \pi_{j_0}^\infty\Big(\bigcap_{j\in\mathbb{Z}^2_\leq}(\pi_j^\infty)^{-1}(C_j)\Big)\\ &\subseteq \pi_{j_0}^\infty((\pi_{j_0}^\infty)^{-1}(C_{j_0})) = C_{j_0}\end{aligned}$$

Then for any j, $\pi_j^\infty(x,z) = (\pi_j^\infty x,\pi_j^\infty y)\in S_j$ and $(\pi_j^\infty y,\pi_j^\infty z)\in T_j$ implies that $(\pi_j^\infty x,\pi_j^\infty z)\in T_j\circ S_j = (T\circ S)_j$. It follows that $(x,z)\in\mathrm{Lim}(\mathbf{T}\circ\mathbf{S})$.

Preservation of composition: oplaxness. Now we show the (harder) oplaxness $\mathrm{Lim}(\mathbf{T}\circ\mathbf{S})\subseteq\mathrm{Lim}(\mathbf{T})\circ\mathrm{Lim}(\mathbf{S})$. Suppose $(x,z)\in\mathrm{Lim}(\mathbf{T}\circ\mathbf{S})$ then for all j, $(\pi_j^\infty x,\pi_j^\infty z)\in(T\circ S)_j = T_j\circ S_j$ therefore there must be some y_j such that $(\pi_j^\infty x, y)\in S_j$ and $(y_j,\pi_j^\infty z)\in T_j$. Therefore the following sets are non-empty:

$$W_j = \left\{ y_j\in\prod_{i\in I}Y_i \;\middle|\; (\pi_j^\infty x, y_j)\in S_j,\ (y_j,\pi_j^\infty z)\in T_j\right\}$$

They are moreover closed, since W_j is the intersection of the image of $\{\pi_j^\infty z\}$ under S_j and the preimage of $\{\pi_j^\infty z\}$ under T_j and both are closed relations.

Take $j'\supseteq j$, then:

$$\begin{aligned}\pi_j^{j'}(W_{j'}) &= \left\{ y_j\in\prod_{i\in j}Y_i \;\middle|\; \exists y_{j'}\in\prod_{i\in j'}Y_i : \begin{array}{r}\pi_j^{j'}(y_{j'}) = y_j,\\ (\pi_{j'}^\infty x, y_{j'})\in S_{j'},\\ (y_{j'},\pi_{j'}^\infty z)\in T_{j'}\end{array}\right\}\\ &\subseteq \left\{ y_j\in\prod_{i=\ell}^{r}Y_i \;\middle|\; \begin{array}{c}(\pi_j^\infty x, y_j)\in S_j,\\ (y_j,\pi_j^\infty z)\in T_j\end{array}\right\} = W_j.\end{aligned}$$

It follows that $(\pi_j^{j'})^{-1}(W_j)\supseteq W_{j'}$.

Then for each $j\in\mathcal{J}$, define:

$$C_j \coloneqq \left(\pi_j^\infty\right)^{-1}(W_j) \;\subseteq\; \prod_{i\in\mathbb{Z}}Y_i.$$

Whenever $j\subseteq j'$, $\pi_j^\infty = \pi_j^{j'}\circ\pi_{j'}^\infty$, we have

$$(\pi_j^\infty)^{-1} = (\pi_{j'}^\infty)^{-1}\circ(\pi_j^{j'})^{-1},$$

and therefore

$$\begin{aligned}C_j &= (\pi_j^\infty)^{-1}(W_j) = (\pi_{j'}^\infty)^{-1}\circ(\pi_j^{j'})^{-1}(W_j)\\ &\supseteq (\pi_{j'}^\infty)^{-1}(W_{j'}) = C_{j'}\,.\end{aligned}$$

Since each W_j is closed and the projections π_j^∞ are continuous, C_j is also closed. Moreover, since each W_j is non-empty so is C_j.

We show that, since $\mathcal{J}$ is *directed*, the C_j satisfy the finite intersection property. Fix any n and let $\{j_1,\ldots,j_n\}$ be arbitrary contexts, by directedness there exists $j^*\in\mathcal{J}$ such that $j^*\supseteq j_i$ for all i. Then $C_{j^*}\subseteq\bigcap_{i=1}^n C_{j_i}$ and since C_{j^*} is non-empty, the finite intersection property holds.

Now, by Tychonoff's theorem, the intersection of closed subsets of a compact space satisfying the finite intersection property is nonempty, so that:

$$\bigcap_{j\in\mathcal{J}}(\pi_j^\infty)^{-1}(W_j) = \bigcap_{j\in\mathcal{J}}C_j \neq \varnothing$$

Finally, take $y\in\bigcap_{j\in\mathcal{J}}C_j$, then for every j, $\pi_j^\infty(x,y)\in S_j$ and $\pi_j^\infty(y,z)\in T_j$. Therefore $(x,y)\in\mathrm{Lim}(\mathbf{S})$ and $(y,z)\in\mathrm{Lim}(\mathbf{T})$ and it follows that $\mathrm{Lim}(\mathbf{T}\circ\mathbf{S})\subseteq\mathrm{Lim}(\mathbf{T})\circ\mathrm{Lim}(\mathbf{S})$. □

Definition E.1. A poset $(P,\leq)$ is **Artinian** in case for every sequence $(x_k)_{k\in\mathbb{N}}$ in P with $x_{k+1}\leq x_k$ for all $k\in\mathbb{N}$, there exists $d\in\mathbb{N}$ such that $x_k = x_d$ for all $k\geq d$.

Importantly, the example that we care about, $\mathsf{Rel}(\mathsf{FSet})$, is enriched in Artinian posets since every strictly descending chain of subsets of a finite set terminates.

Lemma E.2. *Let C be an Artinian poset-enriched compact-closed discard category satisfying $1_X \subseteq \bot_X \circ \top_X$ and $\top_X \circ \bot_X \subseteq 1_I$. Let $\mathcal{J} \subseteq \mathcal{P}_f(I)$ be upward directed. Then every monotone net S of morphisms in C is observationally equivalent to a unique causal net $\mathtt{fix}(\mathsf{S})$.*

Proof. Fix $j \in \mathcal{J}$. For each $j' \in \mathcal{J}$ with $j \subseteq j'$, set

$$B_j^{j'} \coloneqq (1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ S_{j'} \circ (1_{X_j} \otimes \bot_{X_{j' \setminus j}}) \;:\; X_j \to Y_j.$$

By monotonicity of S, enlarging j' can only refine this morphism, so the family $\{B_j^{j'} \mid j' \in \mathcal{J},\ j \subseteq j'\}$ stabilises at some $\rho(\mathsf{S}, j) \in \mathcal{J}$ with $j \subseteq \rho(\mathsf{S}, j)$. Define

$$\mathtt{fix}(\mathsf{S}) \coloneqq \{\, B_j^{\rho(\mathsf{S},j)} \,\}_{j \in \mathcal{J}}.$$

We show that $\mathtt{fix}(\mathsf{S})$ is causal. Let $j \subseteq j'$. Then

$$(1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ \mathtt{fix}(\mathsf{S})_{j'} = (1_{Y_j} \otimes \top_{Y_{j' \setminus j}}) \circ B_{j'}^{\rho(\mathsf{S},j')} = B_j^{\rho(\mathsf{S},j')}.$$

Since $\rho(\mathsf{S}, j')$ contains j', it contains j, and by stabilisation of the family for j we have $B_j^{\rho(\mathsf{S},j')} = B_j^{\rho(\mathsf{S},j)} = \mathtt{fix}(\mathsf{S})_j$. Thus $\mathtt{fix}(\mathsf{S})$ is causal.

We show that $\mathtt{fix}(\mathsf{S})$ is observationally equivalent to S. For each j, monotonicity at $j \subseteq \rho(\mathsf{S}, j)$ gives

$$(1_{Y_j} \otimes \top_{Y_{\rho(\mathsf{S},j) \setminus j}}) \circ S_{\rho(\mathsf{S},j)} \subseteq S_j \otimes \top_{X_{\rho(\mathsf{S},j) \setminus j}}.$$

Precomposing $1_{X_j} \otimes \bot_{X_{\rho(\mathsf{S},j) \setminus j}}$ and using $\top \circ \bot \subseteq 1_I$ yields $\mathtt{fix}(\mathsf{S})_j \subseteq S_j$, and thus $\mathtt{fix}(\mathsf{S}) \subseteq \mathsf{S}$.

For the converse, using $1 \subseteq \bot \circ \top$ on the extra input factors gives

$$(1_{Y_j} \otimes \top_{Y_{\rho(\mathsf{S},j) \setminus j}}) \circ S_{\rho(\mathsf{S},j)} \subseteq \mathtt{fix}(\mathsf{S})_j \otimes \top_{X_{\rho(\mathsf{S},j) \setminus j}},$$

so $\mathsf{S} \subseteq \mathtt{fix}(\mathsf{S})$. Therefore S and $\mathtt{fix}(\mathsf{S})$ are observationally equivalent.

We show uniqueness. Let $\mathbf{T}$ be a causal net observationally equivalent to S. Then $\mathbf{T}$ is observationally equivalent to $\mathtt{fix}(\mathsf{S})$. Since $\mathbf{T}$ and $\mathtt{fix}(\mathsf{S})$ are causal, approximation implies pointwise comparison, hence $\mathbf{T} = \mathtt{fix}(\mathsf{S})$ componentwise. □

Lemma E.3. *Given a set I and cofinal $\mathcal{J} \subseteq \mathcal{P}_f(I)$, and any causal monotone net $\mathsf{S} = \{S_j\}_{j \in \mathcal{J}}$ in $\mathsf{Rel}(\mathsf{FSet})$, then for all $j \in \mathcal{J}$,*

$$\pi_j^\infty(\mathtt{lim}(\mathsf{S})) = S_j.$$

Proof. The inclusion $\pi_j^\infty(\mathtt{lim}(\mathsf{S})) \subseteq S_j$ is immediate since $\mathtt{lim}(\mathsf{S}) \subseteq (\pi_j^\infty)^{-1}(S_j)$.

For the reverse inclusion, fix $j_0 \in \mathcal{J}$. If $S_{j_0} = \varnothing$ there is nothing to show. Choose $x_{j_0} \in S_{j_0}$. For each $k \in \mathcal{J}$ with $k \supseteq j_0$, set

$$C_k \coloneqq (\pi_k^\infty)^{-1}(S_k) \;\cap\; (\pi_{j_0}^\infty)^{-1}(\{x_{j_0}\}) \subseteq \prod_{i \in I} X_i.$$

Since S is causal, for every $k \supseteq j_0$ we have $\pi_{j_0}^k(S_k) = S_{j_0}$, so each C_k is nonempty.

Fix $n \in \mathbb{N}$ and a family $(k_m)_{1 \le m \le n}$ in $\mathcal{J}$ with $k_m \supseteq j_0$ for all m. Choose $k^* \in \mathcal{J}$ with $k^* \supseteq \bigcup_{1 \le m \le n} k_m$. Then $C_{k^*} \subseteq \bigcap_{1 \le m \le n} C_{k_m}$, so $\{C_k\}_{k \supseteq j_0}$ has the finite intersection property. Since $\prod_{i \in I} X_i$ is compact Hausdorff, $\bigcap_{k \supseteq j_0} C_k \neq \varnothing$. Choose y in this intersection. Then $y \in \mathtt{lim}(\mathsf{S})$ and $\pi_{j_0}^\infty(y) = x_{j_0}$, so $x_{j_0} \in \pi_{j_0}^\infty(\mathtt{lim}(\mathsf{S}))$. □

From these two lemmas it follows immediate that:

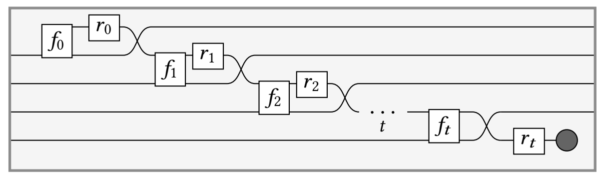
F Proofs of Section 4

Theorem 4.4. *If C is a preorder-enriched discard category satisfying lax-naturality then there is a symmetric monoidal functor $\mathrm{St}([\mathbb{N}, C], \Delta) \to \mathrm{Obs}_{\mathbb{N}}(C)$.*

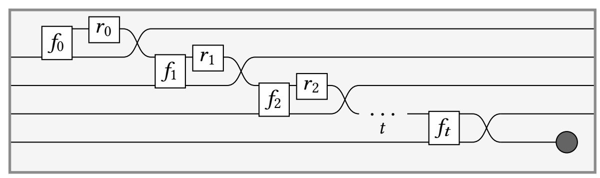
Proof of Theorem 4.4. The functor $U : \mathrm{St}([\mathbb{N}, C], \Delta) \to \mathrm{Obs}_{\mathbb{N}}(C)$ is defined on a representative stateful morphism sequence by unrolling and discarding the output memory. By lax naturality, it is easy to show that this gives a monotone sequence and thus a valid initialised behaviour. To show that two representatives of the same equivalence class of $\mathrm{St}([\mathbb{N}, C], \Delta)$ yield the same initialised behaviour, the only non-trivial thing to prove is that the sliding equation holds. This is shown to follow from lax naturality by the following derivations, where the grey node denotes the discard:

$$U\Big(\mathsf{fbk}^{\mathsf{S}}\big((\mathbf{r} \otimes 1_{\mathsf{Y}}) \circ \mathbf{f}\big)\Big)_{[0,t]}$$

$$=$$



$$\supseteq$$

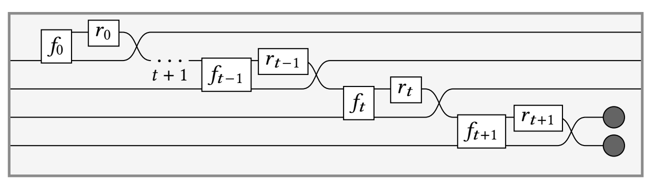


$$=$$

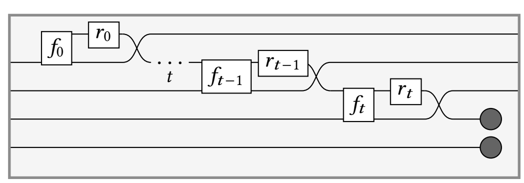
$$U\Big(\mathsf{fbk}^{\mathsf{S}}\big(\mathbf{f} \circ (\mathtt{Shft}(\mathbf{r}) \otimes 1_{\mathsf{X}})\big)\Big)_{[0,t]}$$

$$(1_{Y_{0,t}} \otimes \top_{Y_{t+1}}) \circ \Big(U\Big(\mathsf{fbk}^{\mathsf{S}}\big(\mathbf{f} \circ (\mathtt{Shft}(\mathbf{r}) \otimes 1_{\mathsf{X}})\big)\Big)\Big)_{[0,t+1]}$$

$$=$$



$$\supseteq$$



$$=$$

$$U\Big(\mathsf{fbk}^{\mathsf{S}}\big((\mathbf{r} \otimes 1_{\mathsf{Y}}) \circ \mathbf{f}\big)\Big)_{[0,t]} \otimes \top_{X_{t+1}}$$

Note that the second inequality requires looking ahead to the context $[0, t+1] \supseteq [0, t]$. □

Proposition 4.5. *Given a partial Markov category C with the conditional preorder, $U : \mathrm{St}([\mathbb{N}, C], \Delta) \to \mathrm{Obs}_{\mathbb{N}}(C)$ is full.*

Proof of Proposition 4.5. Given an initialised behaviour in $\mathrm{Obs}_{\mathbb{N}}(C)$, any representative of the equivalence class is a monotone sequence in $\mathsf{Mon}_{\mathbb{N}}(C)$.

Take a representative monotone sequence $f : \mathbf{X} \to \mathbf{Y}$, then we have

$$(1_{Y_{[0,t]}} \otimes \top_{Y_{t+1}}) \circ f_{t+1} \subseteq f_t \otimes \top_{X_{t+1}}$$

By definition of the conditional preorder, there exists an effect $r_t : Y_{[0,t]} \otimes X_{[0,t+1]} \to I$ such that

$$\begin{aligned}&(1_{Y_{[0,t]}} \otimes \top_{Y_{t+1}}) \circ f_{t+1}\\ &\quad = (1_{Y_{[0,t]}} \otimes r_t) \circ ((\delta_{Y_{[0,t]}} \circ f_t) \otimes 1_{X_{[0,t+1]}}) \circ (\delta_{X_{[0,t]}} \otimes 1_{X_{t+1}})\end{aligned}$$

Moreover, because f_{t+1} has a conditional in C, there exists $h_t : Y_{[0,t]} \otimes X_{[0,t+1]} \to Y_{t+1}$ such that:

$$f_{t+1} = (1_{Y_{[0,t]}} \otimes h_t) \circ ((\delta_{Y_{[0,t]}} \otimes \top_{Y_{t+1}}) \circ f_{t+1}) \otimes 1_{X_{[0,t+1]}}) \circ \delta_{X_{[0,t+1]}}$$

Plugging these two equations together we obtain:

$$f_{t+1} = (1_{Y_{[0,t]}} \otimes c_t) \circ ((\delta_{Y_{[0,t]}} \circ f_t) \otimes 1_{X_{[0,t+1]}}) \circ (\delta_{X_{[0,t]}} \otimes 1_{X_{t+1}}) \quad (8)$$

where $c_t = (h_t \otimes r_t) \circ \delta_{Y_{[0,t]} \otimes X_{[0,t+1]}}$.

Now define a sequence $g \in [\mathbb{N}, C]$ by $g_0 = (1_{X_0} \otimes (\delta_{Y_0} \circ f_0)) \circ \delta_{X_0}$ and

$$\begin{aligned}g_{t+1} = &(1_{Y_{[0,t]}} \otimes \mathsf{swap}_{X_{[0,t+1]}, Y_{t+1}} \otimes 1_{Y_{t+1}})\\ &\circ (1_{Y_{[0,t]} \otimes X_{[0,t+1]}} \otimes (\delta_{Y_{t+1}} \circ c_t)) \circ \delta_{Y_{[0,t]} \otimes X_{[0,t+1]}}\end{aligned}$$

This defines a stateful morphism sequence $g : \mathbf{X} \to \mathbf{Y} \in \mathrm{St}([\mathbb{N}, C])$ with memory object at time step t given by $X_{0,t} \otimes Y_{0,t}$, and by Equation (8) we have $U(g) = f$. □

Proposition 4.6. *Given a partial Markov category C with ranges and the conditional preorder enrichment, let $\mathsf{Tot}(C)$ be the full subcategory of total morphisms, then $\mathsf{Stream}_{\mathsf{Tot}(C)}(C) \simeq \mathsf{Mon}_{\mathbb{N}}(C)$ and it follows that there is a full functor $\mathsf{Stream}_{\mathsf{Tot}(C)}(C) \to \mathrm{Obs}_{\mathbb{N}}(C)$.*

Proof of Proposition 4.6. Note that first the notion of causal sequence in the sense of Bonchi et al. [6, Def 6.1] is precisely a monotone sequence in $\mathsf{Mon}_{\mathbb{N}}(C)$ [6, Prop. 6.7]. Therefore [6, Thm 7.19] implies precisely that $\mathsf{Stream}_{\mathsf{Tot}(C)}(C) \simeq \mathsf{Mon}_{\mathbb{N}}(C)$ when C has ranges. Moreover, the functor $\mathsf{Mon}_{\mathbb{N}}(C) \to \mathrm{Obs}_{\mathbb{N}}(C)$ induced by the observational equivalence quotient is clearly full, concluding the proof. □

Lemma 4.9. *If C is a preorder-enriched compact-closed discard category satisfying lax naturality, then for any object X of C:*

$$\top_X \circ \bot_X \subseteq 1_I \text{ and } 1_X \subseteq \bot_X \circ \top_X$$

Proof of Lemma 4.9. The first inequation follows directly by lax-naturality of $\top_X$. The second inequation follows by the snake equation and lax naturality:

$$\begin{aligned}1_X &= (1_X \otimes \cap_X) \circ (\cup_X \otimes 1_X)\\ &\subseteq (1_X \otimes \top_X \otimes \top_X) \circ (\cup_X \otimes 1_X)\\ &= \bot_X \circ \top_X\end{aligned}$$

□

Theorem 4.10. *If C is a preorder-enriched compact-closed discard category satisfying lax naturality, then there is a monoidal natural transformation*

$$\delta : 1_{\mathrm{Obs}_{\mathbb{Z}}(C)} \Rightarrow \mathtt{Shft}; \quad \delta_{\mathbf{X}} := \Big[\Big\{\bot_{X_{\ell-1}} \otimes \Big(\bigotimes_{k=\ell}^{r-1} 1_{X_k}\Big) \otimes \top_{X_r}\Big\}_{[\ell,r] \in \mathbb{Z}^2_{\leq}}\Big],$$

making $\mathrm{Obs}_{\mathbb{Z}}(C)$ into a compact-closed feedback category.

Proof of Theorem 4.10. Following Proposition 2.10, it is sufficient to show that $\mathrm{Obs}_{\mathbb{Z}}(C)$ is compact-closed and that δ is a monoidal natural transformation.

Compact closure. For every object $\mathbf{X}$ of $\mathrm{Obs}_{\mathbb{Z}}(C)$, we define the dual as $(\mathbf{X})^* := (X_k^*)_{k \in \mathbb{Z}}$. The cups and caps are obtained component-wise by:

$$\cup_{\mathbf{X}} := \left[\left\{(\beta_{\mathbf{X}^*, \mathbf{X}})^{-1}_{[\ell,r]} \circ \bigotimes_{k=\ell}^{r} \cup_{X_k}\right\}_{[\ell,r] \in \mathbb{Z}^2_{\leq}}\right],$$
$$\cap_{\mathbf{X}} := \left[\left\{\bigotimes_{k=\ell}^{r} \cap_{X_k} \circ (\beta_{\mathbf{X}, \mathbf{X}^*})_{[\ell,r]}\right\}_{[\ell,r] \in \mathbb{Z}^2_{\leq}}\right].$$

Then lax naturality ensures that $\cap_{X_k} \subseteq \top_{X_k^* \otimes X_k}$ and $\top_{X_k \otimes X_k^*} \circ \cup_{X_k} \subseteq 1_I = \top_I$. It follows that $\cup_{\mathbf{X}}$ and $\cap_{\mathbf{X}}$ are monotone sequences. The snake equations can be shown component-wise.

Monotonicity of delay. We start by showing monotonicity of each component of δ. Note that for any $[\ell, r] \in \mathbb{Z}^2_{\leq}$,

$$\begin{aligned}(\top_{X_{\ell-2}} \otimes 1_{X_{[\ell-1,r-1]}}) \circ (\delta_{\mathbf{X}})_{[\ell-1,r]} &= 1_{X_{[\ell-1,r-1]}} \otimes \top_{X_r}\\ &\subseteq \top_{X_{\ell-1}} \otimes (\bot_{X_{\ell-1}} \otimes 1_{X_{[\ell-1,r-1]}} \otimes \top_{X_r})\\ &= \top_{X_{\ell-1}} \otimes (\delta_{\mathbf{X}})_{[\ell,r]}\end{aligned}$$

where the inequality follows from $1_{X_{\ell-1}} \subseteq \bot_{X_{\ell-1}} \circ \top_{X_{\ell-1}}$. On the other hand we have:

$$(1_{X_{[\ell,r]}} \otimes \top_{X_{r+1}}) \circ (\delta_{\mathbf{X}})_{[\ell,r+1]} = (\delta_{\mathbf{X}})_{[\ell,r]} \otimes \top_{X_{r+1}}$$

Therefore $\delta_{\mathbf{X}}$ is a monotone sequence.

Monoidal naturality of delay. To show naturality, fix any $\mathbf{f} : \mathbf{X} \to \mathbf{Y}$ and consider the compositions $\mathtt{Shft}(\mathbf{f}) \circ \delta_{\mathbf{X}} : \mathbf{X} \to \mathtt{Shft}(\mathbf{Y})$ and $\delta_{\mathbf{Y}} \circ \mathbf{f} : \mathbf{X} \to \mathtt{Shft}(\mathbf{Y})$. Fix any $[\ell, r] \in \mathbb{Z}^2_{\leq}$. First, to show $\mathtt{Shft}(\mathbf{f}) \circ \delta_{\mathbf{X}} \subseteq \delta_{\mathbf{Y}} \circ \mathbf{f}$ we look ahead to the context $[l, r+1] \supseteq [\ell, r]$:

$$\begin{aligned}&(1_{Y_{[\ell-1,r-1]}} \otimes \top_{Y_r}) \circ (\mathtt{Shft}(\mathbf{f}) \circ \delta_{\mathbf{X}})_{[\ell,r+1]}\\ &\subseteq ((\bot_{Y_{\ell-1}} \circ \top_{Y_{\ell-1}}) \otimes 1_{Y_{[\ell,r-1]}} \otimes \top_{Y_r}) \circ f_{[l-1,r]} \circ (\delta_{\mathbf{X}})_{[\ell,r+1]}\\ &\subseteq (\bot_{Y_{\ell-1}} \otimes 1_{Y_{[\ell,r-1]}} \otimes \top_{Y_r}) \circ f_{[l,r]} \circ (\top_{X_{\ell-1}} \otimes 1_{X_{[\ell,r]}}) \circ (\delta_{\mathbf{X}})_{[\ell,r+1]}\\ &\subseteq (\bot_{Y_{\ell-1}} \otimes 1_{Y_{[\ell,r-1]}} \otimes \top_{Y_r}) \circ f_{[l,r]} \circ (1_{X_{[\ell,r]}} \otimes \top_{X_{r+1}})\\ &= (\delta_{\mathbf{Y}} \circ \mathbf{f})_{[\ell,r]} \otimes \top_{X_{r+1}}\end{aligned}$$

where the first inequality uses $1_{Y_{\ell-1}} \subseteq \bot_{Y_{\ell-1}} \circ \top_{Y_{\ell-1}}$, the second is left monotonicity of f, and the third follows by definition $\delta_{\mathbf{X}}$ and $\top_{X_{r+1}} \circ \bot_{X_{r+1}} \subseteq 1_I$. Similarly, for the other direction we use the context $[l-1,r] \supseteq [\ell,r]$:

$$\begin{aligned}
&(\top_{Y_{\ell-2}} \otimes 1_{Y_{[\ell-1,r-1]}})(\delta_{\mathrm{Y}} \circ \mathbf{f})_{[\ell-1,r]} \\
&\subseteq (1_{Y_{[\ell-1,r-1]}} \otimes \top_{Y_r}) \circ f_{[\ell-1,r]} \\
&\subseteq f_{[\ell-1,r-1]} \otimes \top_{X_r} \\
&\subseteq f_{[\ell-1,r-1]} \circ ((\bot_{X_{\ell-1}} \circ \top_{X_{\ell-1}}) \otimes 1_{X_{[\ell,r-1]}} \otimes \top_{X_r}) \\
&= \top_{X_{\ell-1}} \otimes (f_{[\ell-1,r-1]} \circ (\delta_{\mathrm{X}})_{[\ell,r]}) \\
&= \top_{X_{\ell-1}} \otimes [\mathtt{Shft}(\mathbf{f}) \circ \delta_{\mathrm{X}}]_{[\ell,r]}
\end{aligned}$$

Therefore we have shown that the two sides of naturality are observationally equal.

Finally, coherence of the delay natural transformation with respect to the unit and the monoidal product follows from the coherence axioms of $\top$. $\square$

Corollary 4.11. *Assume C is a preorder-enriched compact-closed discard category satisfying lax naturality. Then there are feedback functors:*

$$\mathrm{St}(C) \to \mathrm{St}([\mathbb{Z}, C], \mathtt{Shft}) \to \mathrm{Obs}_{\mathbb{Z}}(C)$$

Proof of Corollary 4.11. The first functor is given by Lemma 2.12.

By Theorem 4.10, $\mathrm{Obs}_{\mathbb{Z}}(C)$ is a compact-closed feedback category, with delay endofunctor

$$\mathtt{Shft} : \mathrm{Obs}_{\mathbb{Z}}(C) \to \mathrm{Obs}_{\mathbb{Z}}(C).$$

Define a symmetric monoidal functor $J : [\mathbb{Z}, C] \to \mathrm{Obs}_{\mathbb{Z}}(C)$ acting as the identity on objects and on morphisms by:

$$J(f) \coloneqq \Big[\big\{ \textstyle\bigotimes_{k=\ell}^{r} f_k \big\}_{[\ell,r] \in \mathbb{Z}^2_{\leq}} \Big].$$

First we check that the displayed family of contexts defines a monotone sequence. Fix a sequence $f_k : X_k \to Y_k$ and a context $[\ell,r] \in \mathbb{Z}^2_{\leq}$. Using monoidality of $\top$ and $\bot$, the right monotonicity inequality for $J(f)$ reduces to

$$\top_Y \circ f_k \subseteq \top_X,$$

and the left monotonicity inequality reduces to

$$f_k \circ \bot_X \subseteq \bot_Y,$$

both of which hold by lax naturality.

Moreover, it is immediate that J preserves the $\mathtt{Shft}$ endofunctor, i.e. $\mathtt{Shft} \circ J = J \circ \mathtt{Shft}$.

Now apply Proposition C.2 with $\Gamma = 1_{[\mathbb{Z},C]}$ and $\Delta = \mathtt{Shft}$. The Δ-compatibility required there is exactly $\mathtt{Shft} \circ J = J \circ \mathtt{Shft}$, which we have just shown. Hence there exists a unique feedback functor $F : \mathrm{St}([\mathbb{Z}, C]) \to \mathrm{Obs}_{\mathbb{Z}}(C)$ such that $F \circ \eta = J$. This gives the claimed action on objects and on morphisms coming from $[\mathbb{Z}, C]$.

Finally, F is a compact-closed feedback functor because both source and target are compact-closed feedback categories with trivial guard, therefore by Proposition 2.10, the feedback operator and the delay determine one another; thus preserving feedback forces preservation of the induced delay and the compact-closed structure. $\square$

G Proofs of Section 6

Theorem 6.2. *There is an oplax normal feedback functor*

$$\mathcal{Z}^* : \mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))} \to \mathrm{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}),$$

given on objects by $n \mapsto n^{\mathbb{Z}}$ and on morphisms $S : n \to m$ by

$$\mathcal{Z}^*(S) \;\coloneqq\; \Big[\Big\{ \big\{ (\zeta_j^n(\mathbf{x}), \zeta_j^m(\mathbf{y})) \;\big|\; (\mathbf{x}, \mathbf{y}) \in S \big\} \Big\}_{j \in \mathbb{Z}^2_{\leq}} \Big],$$

Where oplax and normal means that

$$\mathcal{Z}^*(T \circ S) \subseteq \mathcal{Z}^*(T) \circ \mathcal{Z}^*(S), \qquad \mathcal{Z}^*(1_n) = 1_{n^{\mathbb{Z}}},$$

Proof of Theorem 6.2. First we show that $\mathcal{Z}^*$ produces a monotone sequence of affine relations.

For all $j \in \mathbb{Z}^2_{\leq}$, $(\mathcal{Z}^*(S))_j$ is an $\mathbb{K}$-affine subspace; moreover for all $n, m \in \mathbb{N}$, ζ_j^n and ζ_j^m are $\mathbb{K}$-linear. Furthermore, each $(\mathcal{Z}^*(S))_j$ is defined via the projection of the same infinite affine subspace, so $\mathcal{Z}^*(S)$ produces a causal, and thus monotone sequences of affine relations.

Next we show that $\mathcal{Z}^*$ is oplax. Fix $j \in \mathbb{Z}^2_{\leq}$. If $T \circ S = \varnothing$ then $(\mathcal{Z}^*(T \circ S))_j = \varnothing$ by definition, so the inclusion is vacuous.

Assume otherwise, so that $T \circ S \neq \varnothing$. Take $(u, w) \in (\mathcal{Z}^*(T \circ S))_j$. By definition of $(\mathcal{Z}^*(T \circ S))_j$, there exists some $(\mathbf{x}, \mathbf{z}) \in T \circ S$ such that

$$u = \zeta_j^n(\mathbf{x}) \qquad \text{and} \qquad w = \zeta_j^p(\mathbf{z}).$$

Since $(\mathbf{x}, \mathbf{z}) \in T \circ S$, by definition of relational composition in $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}((z))}$ there exists some $\mathbf{y} \in \mathbb{K}((z))^m$ such that

$$(\mathbf{x}, \mathbf{y}) \in S \qquad \text{and} \qquad (\mathbf{y}, \mathbf{z}) \in T.$$

Since $(\mathbf{x}, \mathbf{y}) \in S$, by definition of $(\mathcal{Z}^*(S))_j$ we have

$$\big(u, \zeta_j^m(\mathbf{y})\big) = \big(\zeta_j^n(\mathbf{x}), \zeta_j^m(\mathbf{y})\big) \in (\mathcal{Z}^*(S))_j,$$

Similarly

$$\big(\zeta_j^m(\mathbf{y}), w\big) = \big(\zeta_j^m(\mathbf{y}), \zeta_j^p(\mathbf{z})\big) \in (\mathcal{Z}^*(T))_j,$$

Therefore, by definition of relational composition in $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{K}}$,

$$(u, w) \in (\mathcal{Z}^*(T))_j \circ (\mathcal{Z}^*(S))_j.$$

Therefore, $\mathcal{Z}^*$ is oplax.

It follows that $\mathcal{Z}^*$ preserves the identity, as well as all of the other symmetric monoidal and feedback structure, since they are defined contextwise. $\square$

Proposition 6.3. *Given a finite field $\mathbb{F}_q$, and a cofinal set $\mathcal{J} \subseteq \mathcal{P}_f(I)$ there is a faithful poset-enriched discard functor*

$$\mathrm{Lim} : \mathrm{Obs}_{I,\mathcal{J}}(\mathsf{AR}^{\mathsf{fd}}_{\mathbb{F}_q}) \to \mathsf{AR}_{\mathbb{F}_q}.$$

Proof of Proposition 6.3. By Theorem 5.5 there is a poset-enriched discard functor $\mathrm{Lim} : \mathrm{Obs}_{I,\mathcal{J}}(\mathsf{Rel}(\mathsf{KHaus})) \longrightarrow \mathsf{Rel}(\mathsf{KHaus})$. Since $\mathbb{F}_q$ is a finite field, any vector space over $\mathbb{F}_q$ is compact and Hausdorff with respect to the product topology discrete topologies. Moreover since the projections are linear maps, inverse images of linear maps are linear maps, and intersections of linear subspaces are linear subspaces, it follows that there is an induced poset enriched discard functor $\mathrm{Obs}_{I,\mathcal{J}}(\mathsf{AR}_{\mathbb{F}_q}) \to \mathsf{AR}_{\mathbb{F}_q}$. Moreover, because the objects of $\mathsf{AR}^{\mathsf{fd}}_{\mathbb{F}_q}$ are compact, by Corollary 5.6, it follows that this functor is moreover faithful. $\square$

Theorem 6.5. *Let $\mathbb{F}_q$ be a finite field and $S \subseteq \mathbb{F}_q(z)^\ell$ be a finite-dimensional $\mathbb{F}_q(z)$-affine subspace. Then*

$$(\mathrm{Lim} \circ \mathcal{Z}^*)(S) = \mathrm{beh}(S) \cap \mathcal{Z}^\gamma_\ell\big(\mathbb{F}_q(z)^\ell\big).$$

Proof of Theorem 6.5. We prove the claim for affine subspaces $S \subseteq \mathbb{F}_q(z)^\ell$; the corresponding statement for affine relations $R \subseteq \mathbb{F}_q(z)^n \times \mathbb{F}_q(z)^m$ follows immediately by viewing R as an affine subspace of $\mathbb{F}_q(z)^{n+m}$ and transporting along the canonical isomorphism

$$(\mathbb{F}_q^{n+m})^{\mathbb{Z}} \cong (\mathbb{F}_q^n)^{\mathbb{Z}} \times (\mathbb{F}_q^m)^{\mathbb{Z}}.$$

If $S = \varnothing$ then $(\mathrm{Lim} \circ \mathcal{Z}^*)(S) = \varnothing = \mathrm{beh}(S)$, so assume $S \neq \varnothing$. Choose $\mathbf{a} \in S$ and put $R \coloneqq S - \mathbf{a}$, so $R \subseteq \mathbb{F}_q(z)^\ell$ is an $\mathbb{F}_q(z)$-linear subspace and $S = R + \mathbf{a}$. By definition of beh on affine subspaces,

$$\mathrm{beh}(S) = \mathrm{beh}(R) + \mathcal{Z}^\gamma_\ell(\mathbf{a}).$$

Since $\mathrm{Lim} \circ \mathcal{Z}^*$ is given on morphisms by $\mathcal{Z}^\gamma$, and $\mathcal{Z}^\gamma$ is $\mathbb{F}_q[z, z^{-1}]$-linear, we also have

$$(\mathrm{Lim} \circ \mathcal{Z}^*)(S) = (\mathrm{Lim} \circ \mathcal{Z}^*)(R) + \mathcal{Z}^\gamma_\ell(\mathbf{a}),$$

and $\mathcal{Z}^\gamma_\ell(\mathbf{a}) \in \mathcal{Z}^\gamma_\ell(\mathbb{F}_q(z)^\ell)$. Thus it suffices to prove the two claims for the linear subspace R.

Choose $r \in \mathbb{N}$ and an $\mathbb{F}_q(z)$-linear map $A : \mathbb{F}_q(z)^\ell \to \mathbb{F}_q(z)^r$ with $R = \ker A$. Pick $d(z) \in \mathbb{F}_q[z, z^{-1}] \setminus \{0\}$ clearing denominators and set $B(z) \coloneqq d(z)A \in \mathbb{F}_q[z, z^{-1}]^{r \times \ell}$. Then $\mathrm{beh}(R) = \ker(B(\Delta)) \subseteq (\mathbb{F}_q^{\mathbb{Z}})^\ell$.

We first show that for all $u \in \mathbb{F}_q((z))^\ell$,

$$\mathcal{Z}^\gamma_r\big(B(z)u\big) = B(\Delta)\mathcal{Z}^\gamma_\ell(u). \tag{9}$$

Since $\mathcal{Z}^\gamma$ is $\mathbb{F}_q[z, z^{-1}]$-linear, for each entry $b(z) \in \mathbb{F}_q[z, z^{-1}]$ and $x \in \mathbb{F}_q((z))$ we have

$$\mathcal{Z}^\gamma_1\big(b(z)x\big) = b(\Delta)\mathcal{Z}^\gamma_1(x),$$

and applying this entrywise to $B(z)$ and componentwise to vectors yields (9).

We now prove $(\mathrm{Lim} \circ \mathcal{Z}^*)(R) \subseteq \mathrm{beh}(R)$. Take $u \in (\mathrm{Lim} \circ \mathcal{Z}^*)(R)$. Then $u = \mathcal{Z}^\gamma_\ell(x)$ for some $x \in R$. Since $x \in R = \ker(B(z))$, we have $B(z)x = 0$, hence

$$0 = \mathcal{Z}^\gamma_r\big(B(z)x\big) = B(\Delta)\mathcal{Z}^\gamma_\ell(x) = B(\Delta)u,$$

so $u \in \ker(B(\Delta)) = \mathrm{beh}(R)$.

We now prove

$$\mathrm{beh}(R) \cap \mathcal{Z}^\gamma_\ell\big(\mathbb{F}_q(z)^\ell\big) \subseteq (\mathrm{Lim} \circ \mathcal{Z}^*)(R).$$

Take $u \in \mathrm{beh}(R) \cap \mathcal{Z}^\gamma_\ell\big(\mathbb{F}_q(z)^\ell\big)$. Choose $x \in \mathbb{F}_q(z)^\ell$ with $u = \mathcal{Z}^\gamma_\ell(x)$. Since $u \in \mathrm{beh}(R) = \ker(B(\Delta))$, we have $B(\Delta)u = 0$, hence

$$0 = B(\Delta)\mathcal{Z}^\gamma_\ell(x) = \mathcal{Z}^\gamma_r\big(B(z)x\big).$$

By injectivity of $\mathcal{Z}^\gamma_r$, $B(z)x = 0$, so $x \in \ker(B(z)) = R$. Thus $u = \mathcal{Z}^\gamma_\ell(x) \in (\mathrm{Lim} \circ \mathcal{Z}^*)(R)$.

Therefore $(\mathrm{Lim} \circ \mathcal{Z}^*)(R) = \mathrm{beh}(R) \cap \mathcal{Z}^\gamma_\ell\big(\mathbb{F}_q(z)^\ell\big)$. □

Lemma 6.7. *There is a full feedback functor $\mathrm{St}_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}) \twoheadrightarrow \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}(z)}$, given by identifying the delays and extending scalars along $\mathbb{K}(z)/\mathbb{K}$, and extending scalars along the geometric series embedding $\mathbb{K}((z))/\mathbb{K}(z)$ induces a feedback functor $\mathtt{Ext} : \mathrm{St}_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}) \to \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}((z))}$.*

Proof of Lemma 6.7. By Proposition 2.10, $\mathrm{St}_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}})$ is presented by freely adding a "discrete-time delay"' automorphism to $\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}$ which is natural on the subcategory $\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}$. □

Lemma 6.8. *Let $\eta : \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}} \to \mathrm{St}_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}})$ be the canonical functor. Define $J : \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}} \to \mathrm{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}})$ by $J(V)_t \coloneqq V$ for all $t \in \mathbb{Z}$, and on morphisms by $J(\varnothing) \coloneqq [\{\varnothing\}_{j \in \mathbb{Z}^2_\leq}]$ and*

$$J(A + \mathbf{a}) \coloneqq \left[\left\{\bigoplus_{t=\ell}^{r} \begin{matrix} A + \mathbf{a} & \text{if } t = 0, \\ A & \text{if } t \neq 0 \end{matrix}\right\}_{[\ell, r] \in \mathbb{Z}^2_\leq}\right].$$

Then there is a unique feedback functor

$$\mathtt{Unroll} : \mathrm{St}_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}) \to \mathrm{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}) \text{ satisfying } \mathtt{Unroll} \circ \eta = J.$$

Proof of Lemma 6.8. We proceed by using the universal property of the state construction (see Proposition C.2).

By inspection, J produces monotone sequences and preserves identities. The only nontrivial part is to show that composition is preserved. Take $f \in \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}(\mathbb{K}^n, \mathbb{K}^m)$ and $g \in \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}(\mathbb{K}^m, \mathbb{K}^o)$. If f or g is empty then $J(g \circ f) = [\{\varnothing\}_{j \in \mathbb{Z}^2_\leq}] = J(g) \circ J(f)$. Similarly if $g \circ f$ is empty, then $J(g \circ f)$ is empty at every context $j \in \mathbb{Z}^2_\leq$, hence by Proposition 3.9, $J(g \circ f) = [\{\varnothing\}_{j \in \mathbb{Z}^2_\leq}] = J(g) \circ J(f)$. Otherwise, suppose that we have nonempty affine relations $A + \mathbf{a} \in \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}(\mathbb{K}^n, \mathbb{K}^m)$ and $B + \mathbf{b} \in \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}(\mathbb{K}^m, \mathbb{K}^o)$, with nonempty composite $C + \mathbf{c} \in \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}(\mathbb{K}^n, \mathbb{K}^o)$. Then:

$$\begin{aligned}
&J(C + \mathbf{c}) \\
&= \left[\left\{\bigoplus_{k=\ell}^{r} \begin{pmatrix} C + \mathbf{c} & \text{if } k = 0 \\ C & \text{if } k \neq 0 \end{pmatrix}\right\}_{[\ell, r] \in \mathbb{Z}^2_\leq}\right] \\
&= \left[\left\{\bigoplus_{k=\ell}^{r} \begin{pmatrix} C + \mathbf{c} & \text{if } k = 0 \\ B \circ A & \text{if } k \neq 0 \end{pmatrix}\right\}_{[\ell, r] \in \mathbb{Z}^2_\leq}\right] \\
&= \left[\left\{\bigoplus_{k=\ell}^{r} \begin{pmatrix} B + \mathbf{b} & \text{if } k = 0 \\ B & \text{if } k \neq 0 \end{pmatrix} \circ \begin{pmatrix} A + \mathbf{a} & \text{if } k = 0 \\ A & \text{if } k \neq 0 \end{pmatrix}\right\}_{[\ell, r] \in \mathbb{Z}^2_\leq}\right] \\
&= \left[\left\{\bigoplus_{k=\ell}^{r} \begin{pmatrix} B + \mathbf{b} & \text{if } k = 0 \\ B & \text{if } k \neq 0 \end{pmatrix}\right\}_{[\ell, r] \in \mathbb{Z}^2_\leq}\right] \\
&\quad \circ \left[\left\{\bigoplus_{k=\ell}^{r} \begin{pmatrix} A + \mathbf{a} & \text{if } k = 0 \\ A & \text{if } k \neq 0 \end{pmatrix}\right\}_{[\ell, r] \in \mathbb{Z}^2_\leq}\right] \\
&= J(B + \mathbf{b}) \circ J(A + \mathbf{a}).
\end{aligned}$$

Let $J|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}$ denote the restriction of J along the inclusion $\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}} \rightarrowtail \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}$. For shift-invariance, take $L \in \mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}(V, W)$ and $[\ell, r] \in \mathbb{Z}^2_\leq$; then

$$\begin{aligned}
\big(\mathtt{Shft}(J|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(L))\big)_{[\ell, r]} &= (J(L))_{[\ell-1, r-1]} = \bigoplus_{t=\ell-1}^{r-1} L \\
&= \bigoplus_{t=\ell}^{r} L = (J(L))_{[\ell, r]}.
\end{aligned}$$

Now apply Proposition C.2 with

$$\Gamma : \mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}} \rightarrowtail \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}, \qquad \Delta = 1_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}, \qquad \Gamma' = 1_{\mathrm{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}})},$$
$$\Delta' = \mathtt{Shft}, \qquad F = J, \qquad F|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}} = J|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}.$$

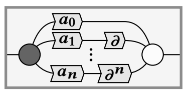
The condition $\Gamma' \circ F|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}} = F \circ \Gamma$ is the statement that $J|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}$ is the restriction of J, and the condition $\Delta' \circ F|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}} = F|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}} \circ \Delta$ is exactly shift-invariance of $J|_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}$. Hence there exists a unique feedback functor

$$\mathtt{Unroll} : \mathsf{St}_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}) \to \mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}})$$

such that $\mathtt{Unroll} \circ \eta = J$. By construction, $\mathtt{Unroll}$ agrees with the defining formulas for J in the statement. □

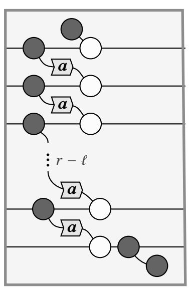
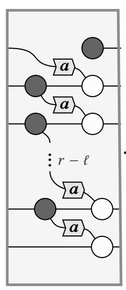
Proposition 6.9. *With respect to polynomials $z + a \in \mathbb{K}[z]$ for $a \neq 0$, $\mathcal{Z}^*(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}((z))})$ satisfies Equation (7) but only satisfies Equation (6) oplaxly.*

Proof of Proposition 6.9. Given a polynomial $a(z) \in \mathbb{K}[\mathbb{Z}]$, define the following morphism in $\mathsf{St}_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}})$:

$\mu_{a(z)} :=$ 

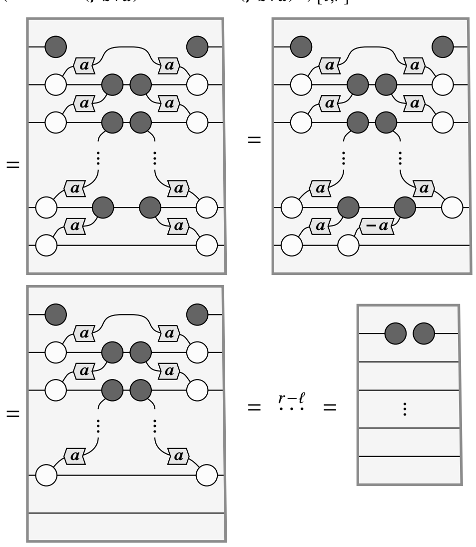
Take some nonzero $a \in \mathbb{K}$.

First, observe that

$\mathtt{Unroll}(\mu_{z+a})_{[\ell,r]}$ =  = .

Therefore

$$(\mathtt{Unroll}(\mu_{z+a}) \circ \mathtt{Unroll}(\mu_{z+a})^*)_{[\ell,r]}$$



From which it follows that

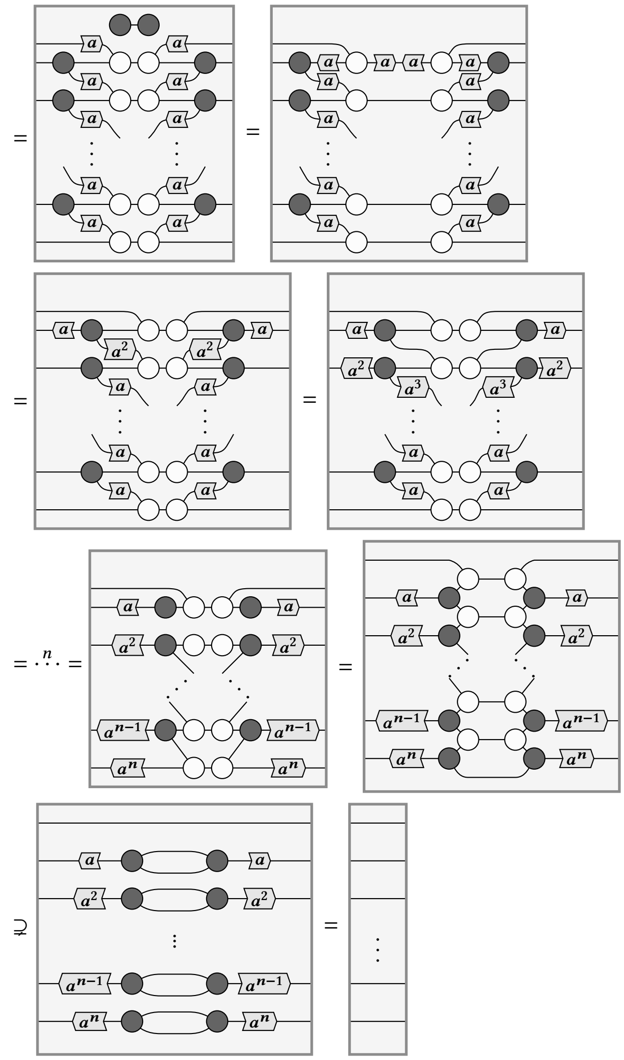
$$\mathtt{Unroll}(\mu_{z+a}) \circ \mathtt{Unroll}(\mu_{z+a})^*))_{[\ell,r]} \circ (\top_1 \oplus 1_{r-\ell}) = \top_1 \oplus 1_{r-\ell},$$

so that

$$\mathtt{Unroll}(\mu_{z+a}) \circ \mathtt{Unroll}(\mu_{z+a})^* = 1_{1\mathbb{Z}}.$$

On the other hand, fixing $n := r - \ell$,

$(\mathtt{Unroll}(\mu_{z+a})^* \circ \mathtt{Unroll}(\mu_{z+a}))_{[\ell,r]}$



Notice that the final inclusion is strict, since the bottom diagram has half the dimension of the one on top. Therefore

$$\mathtt{Unroll}(\mu_{z+a})^* \circ \mathtt{Unroll}(\mu_{z+a}) \supsetneq 1_{1\mathbb{Z}}. \qquad \square$$

Proposition 6.10. *The following diagram laxly commutes:*

$$\begin{tikzcd} \mathsf{St}_{\mathsf{LR}^{\mathrm{fd}}_{\mathbb{K}}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}) \arrow[r, "\mathtt{Unroll}"] \arrow[d, two heads, "\mathtt{Ext}"'] \arrow[dr, phantom, "\cup\!\mathrm{I}"] & \mathsf{Obs}_{\mathbb{Z}}(\mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}}) \\ \mathsf{AR}^{\mathrm{fd}}_{\mathbb{K}((z))} \arrow[ur, bend right, "\mathcal{Z}^*"'] & \end{tikzcd}$$

That is to say, $\mathcal{Z}^*(\mathtt{Ext}(-)) \subseteq \mathtt{Unroll}(-)$.

Proof of Proposition 6.10. A simple diagram chase argument shows that the diagram commutes, respectively on strictly on affine displacements, linear relations and on the delay. Therefore, by oplaxness and normality, it follows that the diagram commutes laxly. □